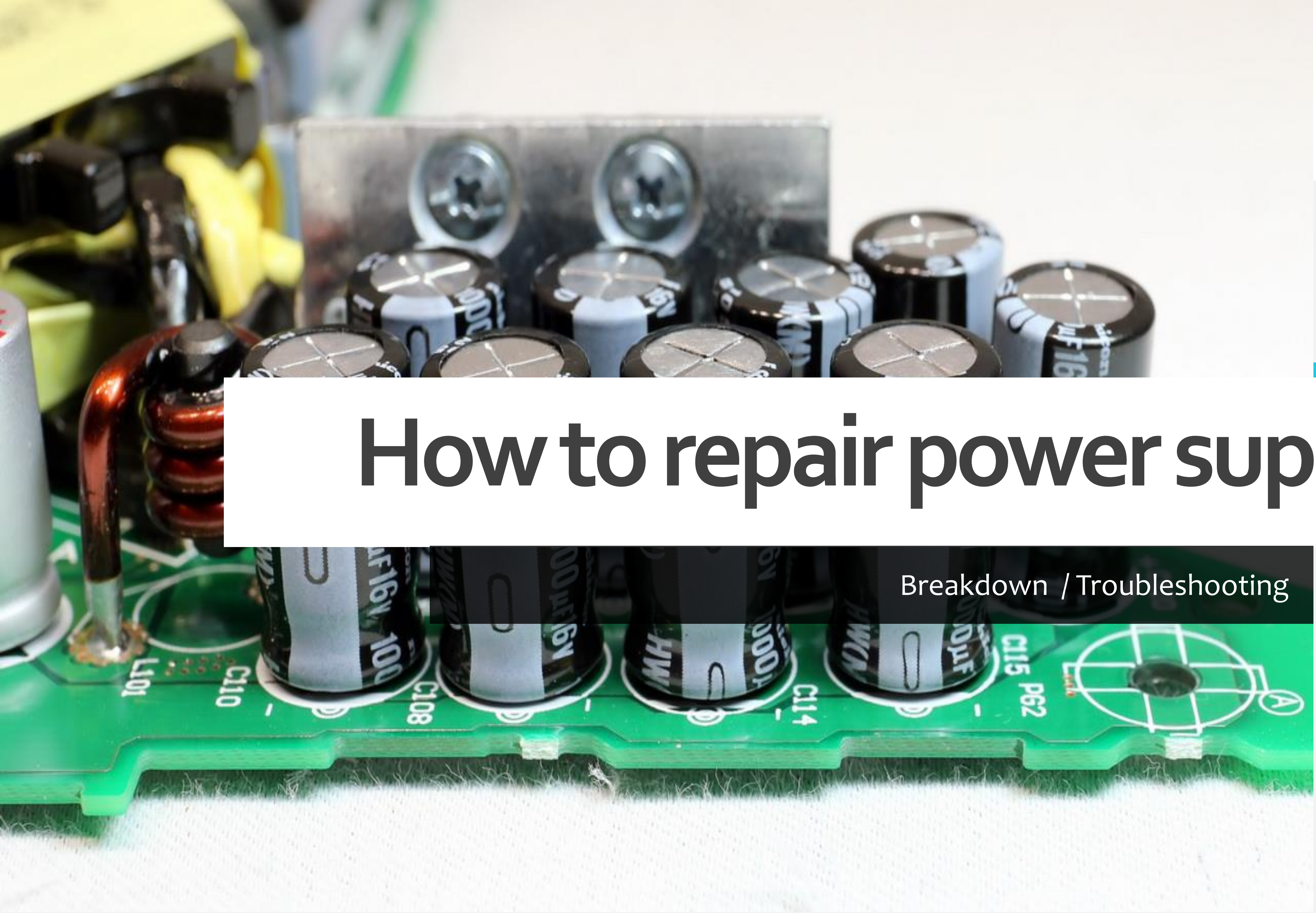


How to repair power supply PS5

Breakdown / Troubleshooting

Which tools do you need when fixing power supplies?

Isolator transformer
When you accidentally touch the hot output you will be save

Variabel transformer
For testing SMPS from low to high AC voltage

A variable transformer is not a Isolator transformer!

Multimeter

Capacitance / ESR meter

To test state of capacitor

Electronic load

To test the secondary output voltage

Oscilloscope

To give a picture of the like PWM signal

Capacitor discharging tool

Safety Guidelines

Electrical Shock

Turn off and unplug power supply when you start repairing.

Wear rubber shows

If you are tired or want to rush things, don't do that and stop!

Best practice is to assume that the power supply is always ON and LIVE

Only use plastic screwdrivers when working on the PCB.

Discharge main capacitor

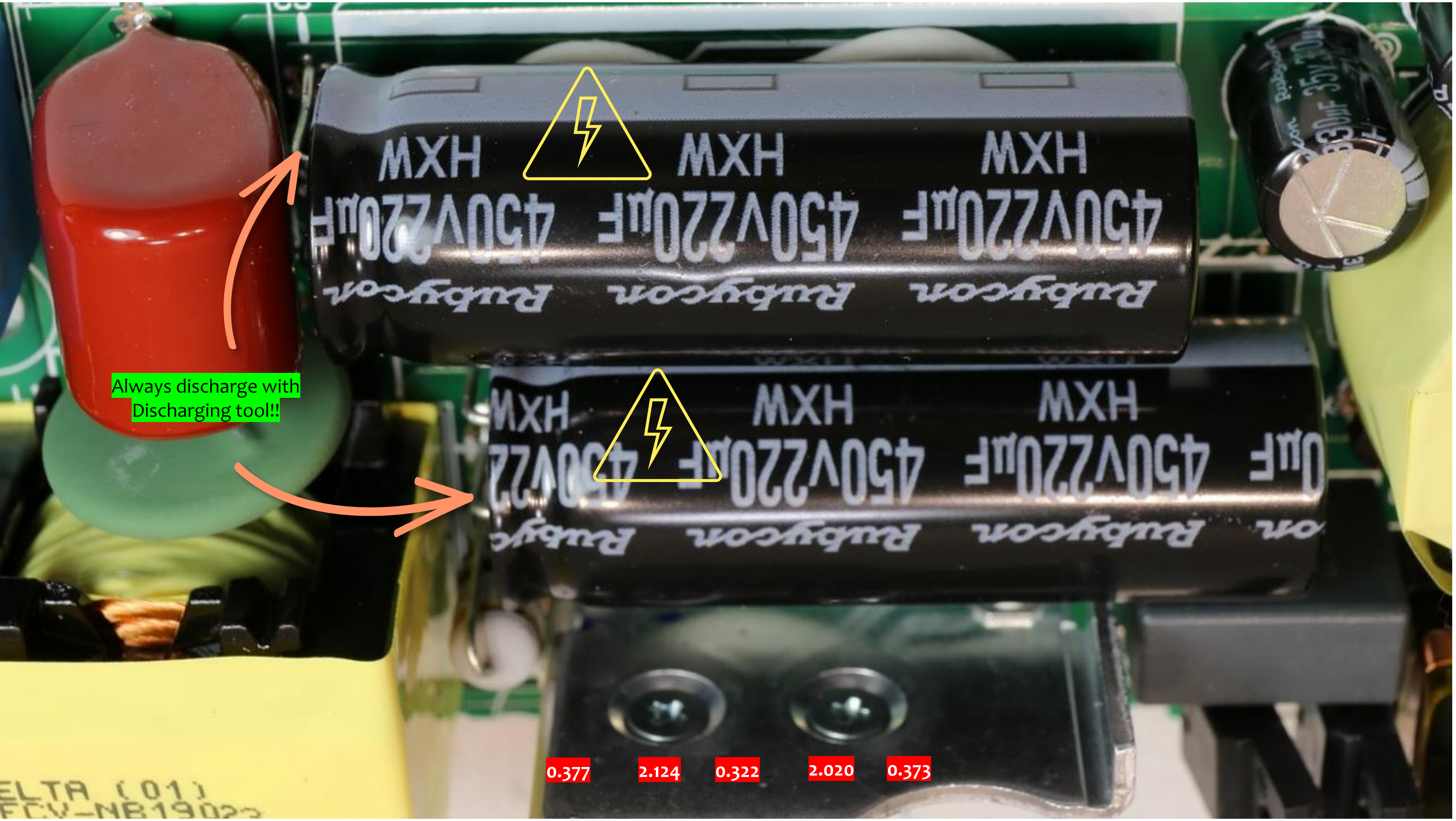
Use a discharging tool to drain the main filter capacitors

Use correct ground

When testing in COLD side use a ground reference on the COLD side.

When testing in HOT side use a ground reference on the HOT side.

These capacitors can kill you!



Disclaimer – Caution!

Power supplies contain capacitors that can store electrical energy even after the supply is disconnected.

Always assume that capacitors are charged and take appropriate precautions to discharge them before touching or working on the circuit. Use suitable tools, such as insulated discharge probes, and follow recommended safety procedures.

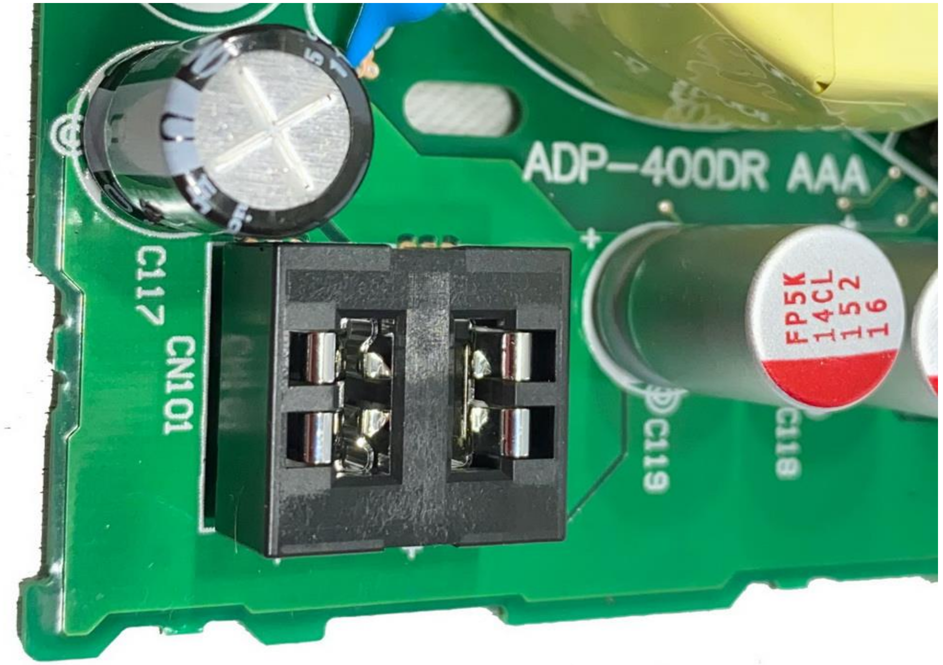
High voltage can be between: 200V - 470V

6 common faults with SMPS

1. No power	Symptom	Test
No power	Fuse blown	RFI /EMI circuit APFC circuit LLC circuit
No power	Fuse good	Hot side Cold side

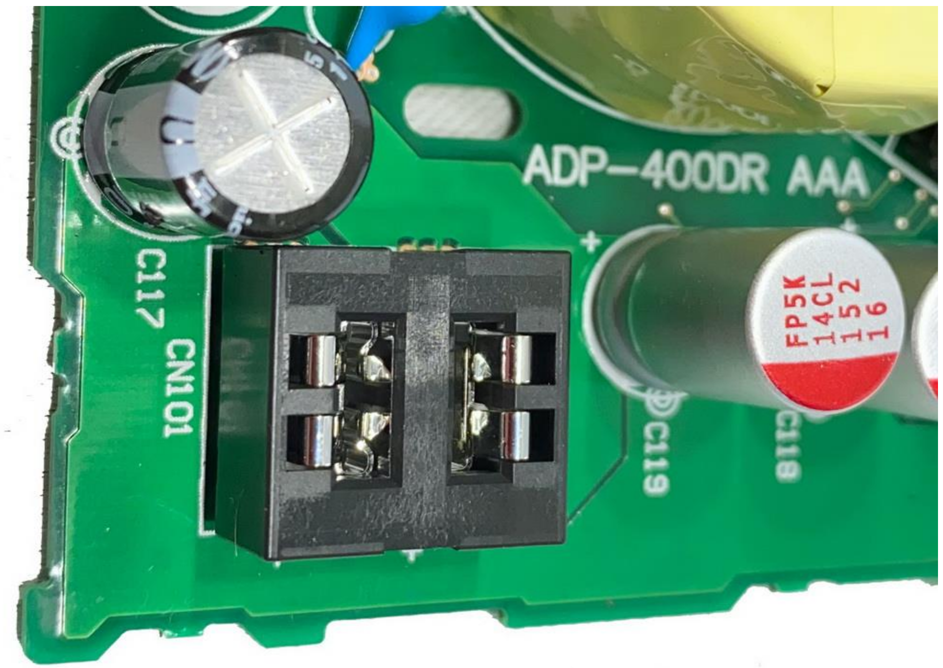
6 common faults with SMPS

2. Low power voltage	Symptom	Test
Output is lower then specified (example: +12V but the output delivers +10V)	No power No standby LED	----



6 common faults with SMPS

3. High power voltage	Symptom	Test
Output is higher then specified (example: +12V but the output delivers +15V)	Power shutdown	----

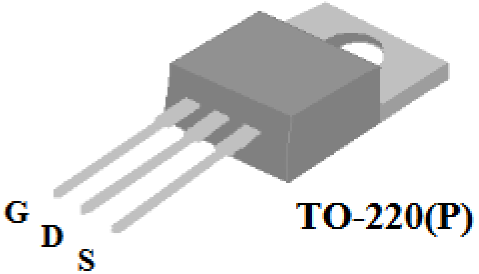


6 common faults with SMPS

4. Power cycling	Symptom	Cause	Test
Output voltage is present then it is turned off and this cycle repeats	The rectifier diodes in the secondary section wil produce an voltage that goes high and then low (on / off)	Bad component on secondary side Bad filter capacitors Faulty component in regulation circuit Error detection / feedback circuit Bad component on primary side	Multimeter red proble to cold ground. Red probe to the output of the rectifier diode.

Measure output voltage on the secondary filter capacitor

PS5 uses fast switching mosfets Instead of diodes.

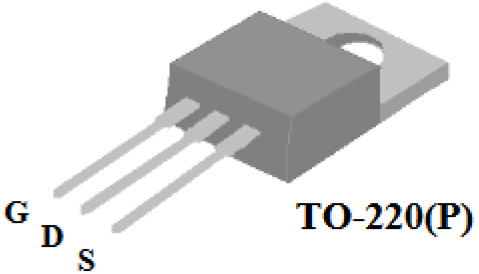


6 common faults with SMPS

5. Power shutdown	Symptom	Cause	Test
When the generated output voltage is too high the SMPS will go in shutdown mode. The SMPS does this too protect the device that is connected to the ouput. In this case the PS5	The rectifier diodes in the secondary section wil produce an voltage that goes to +12V and higher to +20V and then to +0V.	-	Multimeter red proble to cold ground. Red probe to the output of the rectifier diode.

Measure output voltage on the secondary filter capacitor

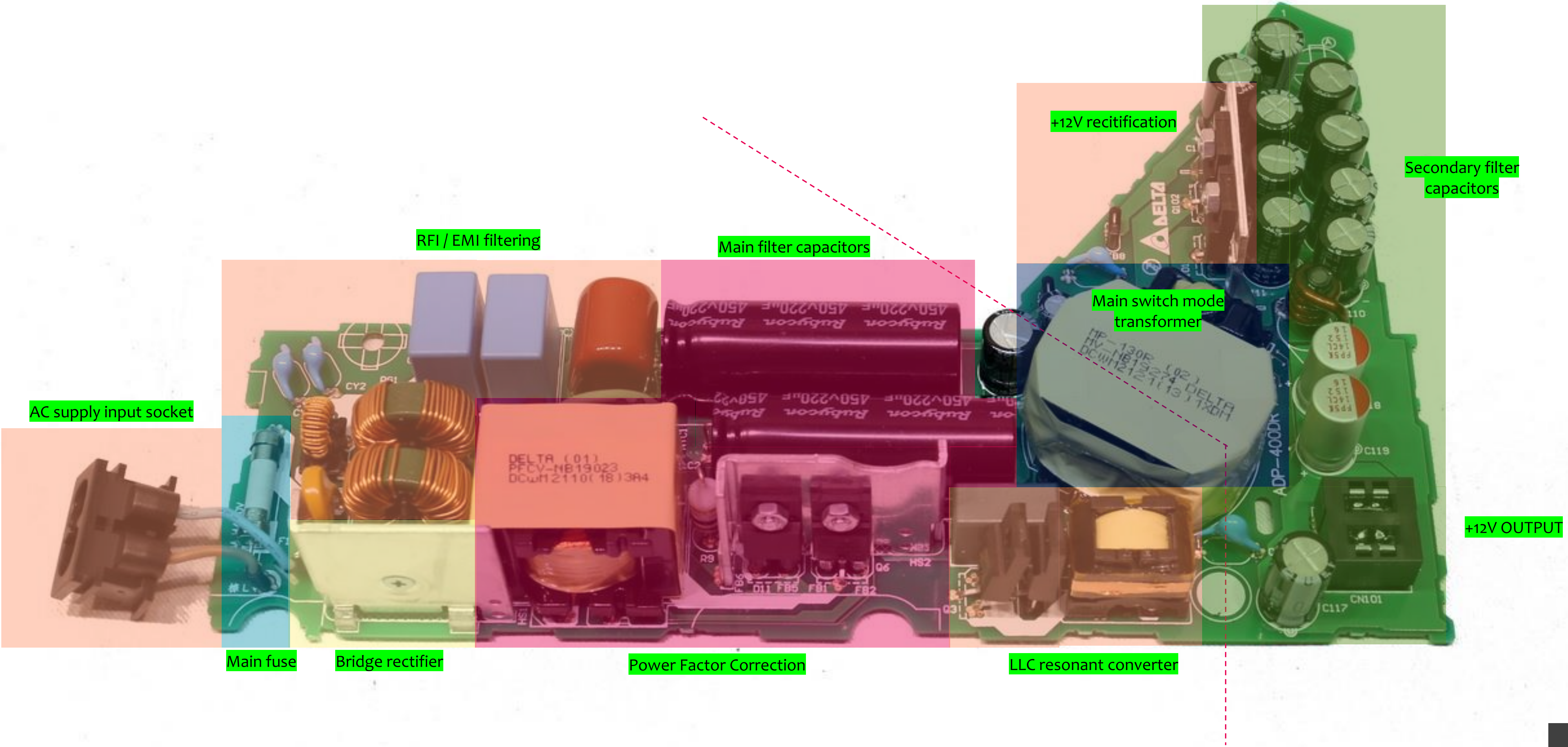
PS5 uses fast switching mosfets Instead of diodes.



6 common faults with SMPS

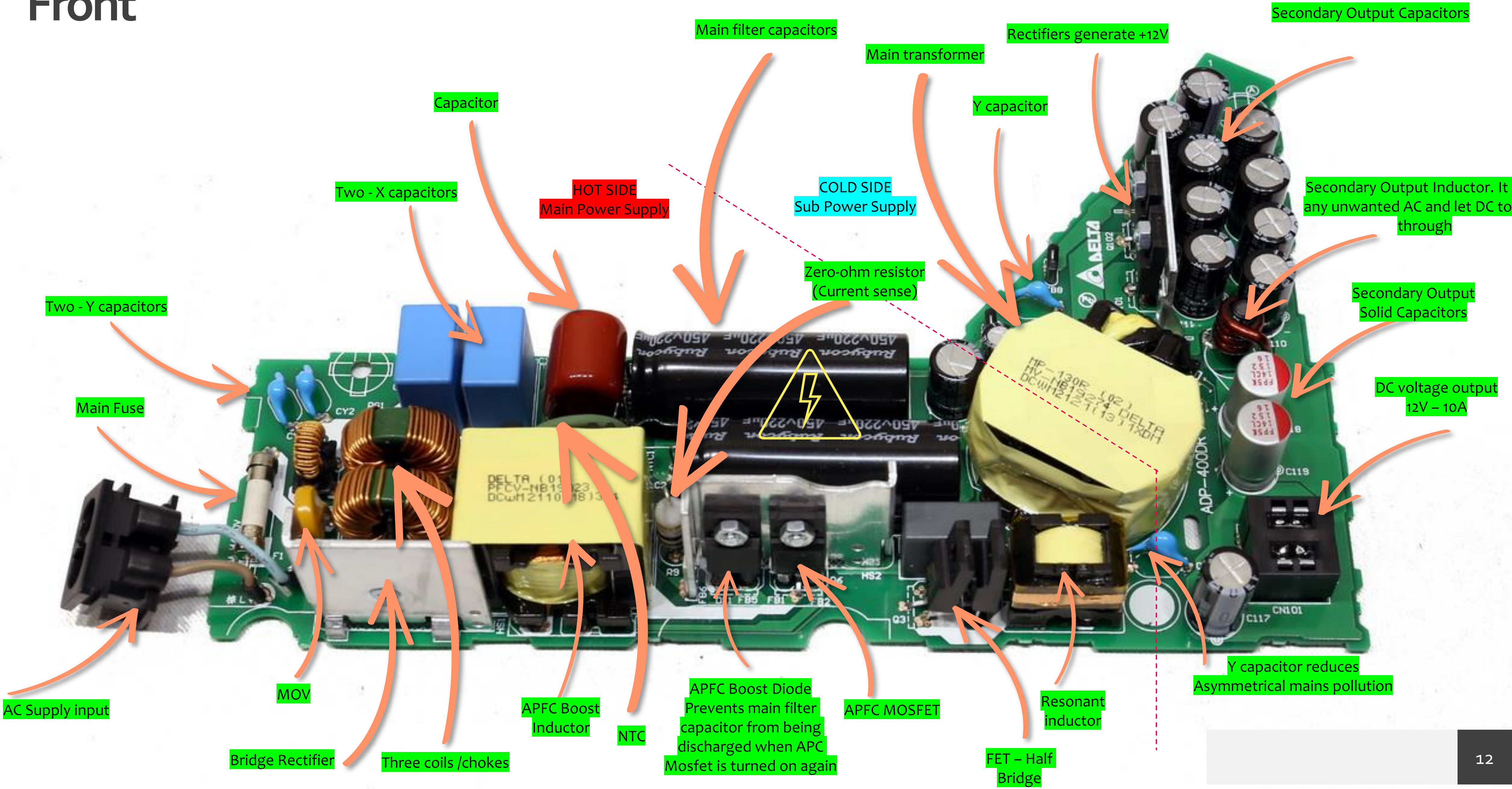
6. Intermittent power	Symptom	Cause	Test
The power supply sometimes works ok and sometimes not.	No power Power	Dry solder joint Loose connection Primary side - startup resistor	Visual inspection Measure startup resistor

Front



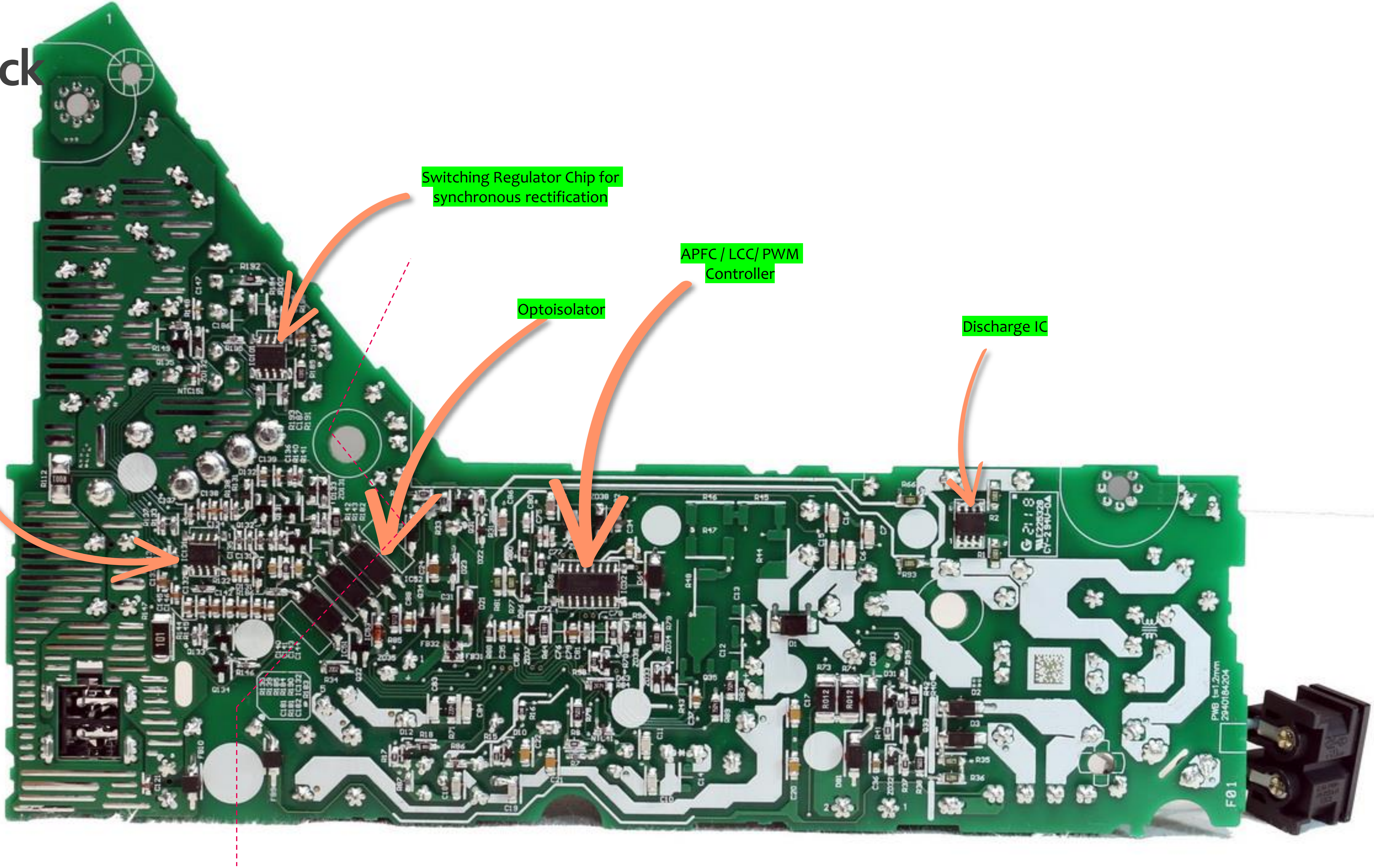
+12V recitification
Secondary filter capacitors
RFI / EMI filtering
Main filter capacitors
Main switch mode transformer
AC supply input socket
+12V OUTPUT
Main fuse
Bridge rectifier
Power Factor Correction
LLC resonant converter

Front



Secondary Output Capacitors
Main filter capacitors
Rectifiers generate +12V
Main transformer
Capacitor
Y capacitor
Two - X capacitors
HOT SIDE
Main Power Supply
COLD SIDE
Sub Power Supply
Secondary Output Inductor. It any unwanted AC and let DC to through
Zero-ohm resistor
(Current sense)
Secondary Output
Solid Capacitors
Two - Y capacitors
DC voltage output
12V – 10A
Main Fuse
Y capacitor reduces
Asymmetrical mains pollution
AC Supply input
MOV
APFC Boost
Inductor
NTC
APFC Boost Diode
Prevents main filter
capacitor from being
discharged when APC
Mosfet is turned on again
APFC MOSFET
Resonant
inductor
Bridge Rectifier
Three coils /chokes
FET – Half
Bridge

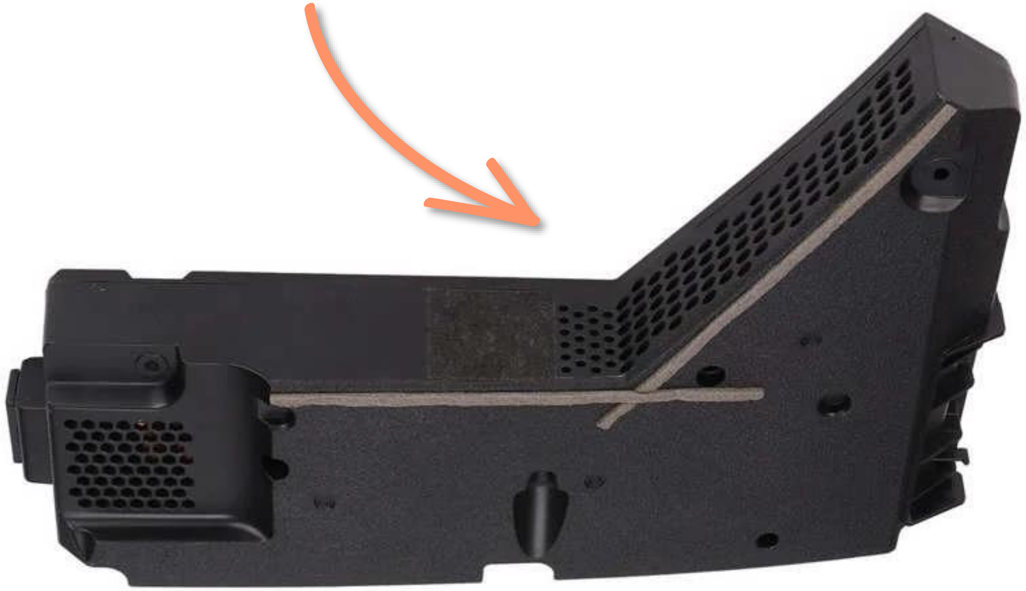
Back



Switching Regulator Chip for synchronous rectification
APFC / LCC/ PWM Controller
Optoisolator
Discharge IC
This is a dual operational amplifier and voltage reference.

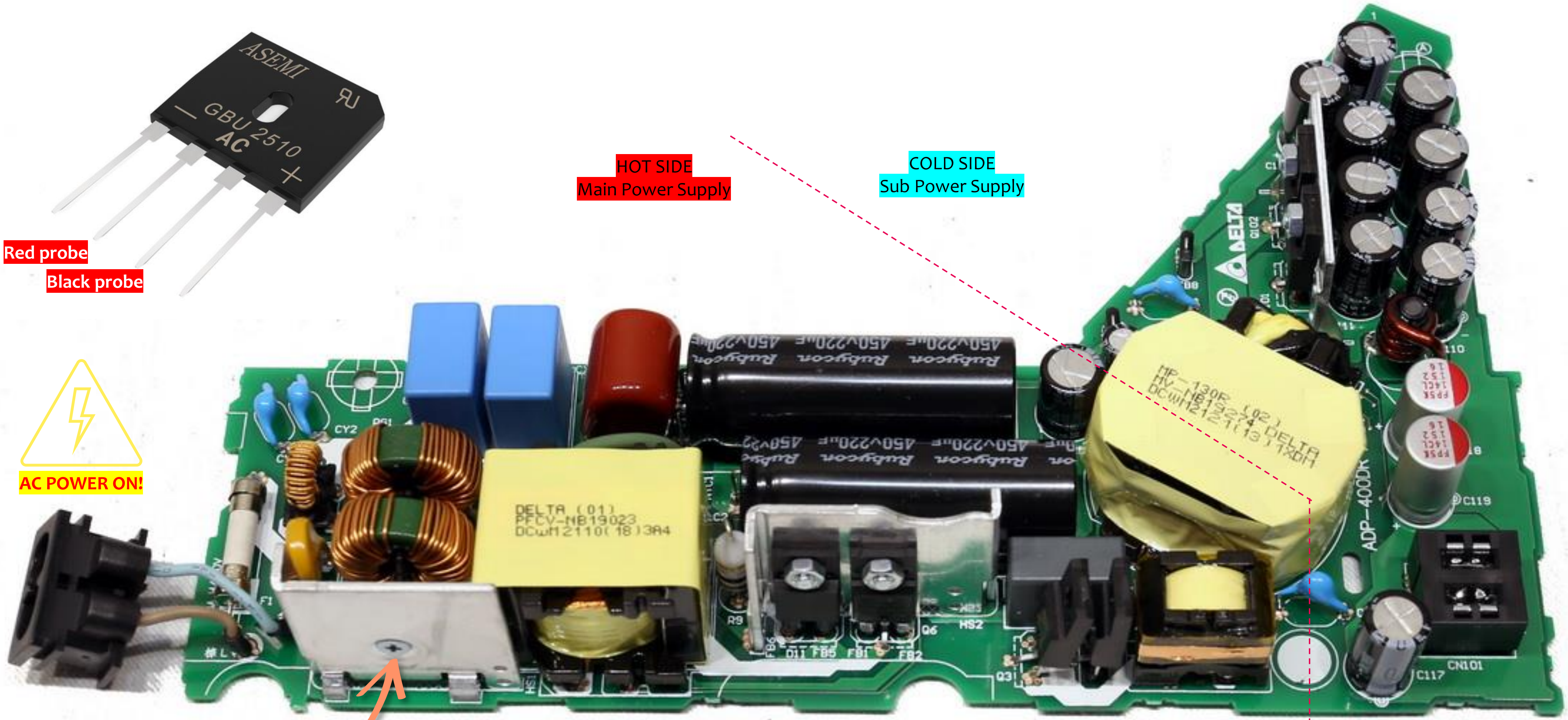
Check inside of case

Search for scorch marks inside of case



Scorch mark

Voltage Testing – 1.1 AC input voltage



Red probe

Black probe

HOT SIDE
Main Power Supply

COLD SIDE
Sub Power Supply

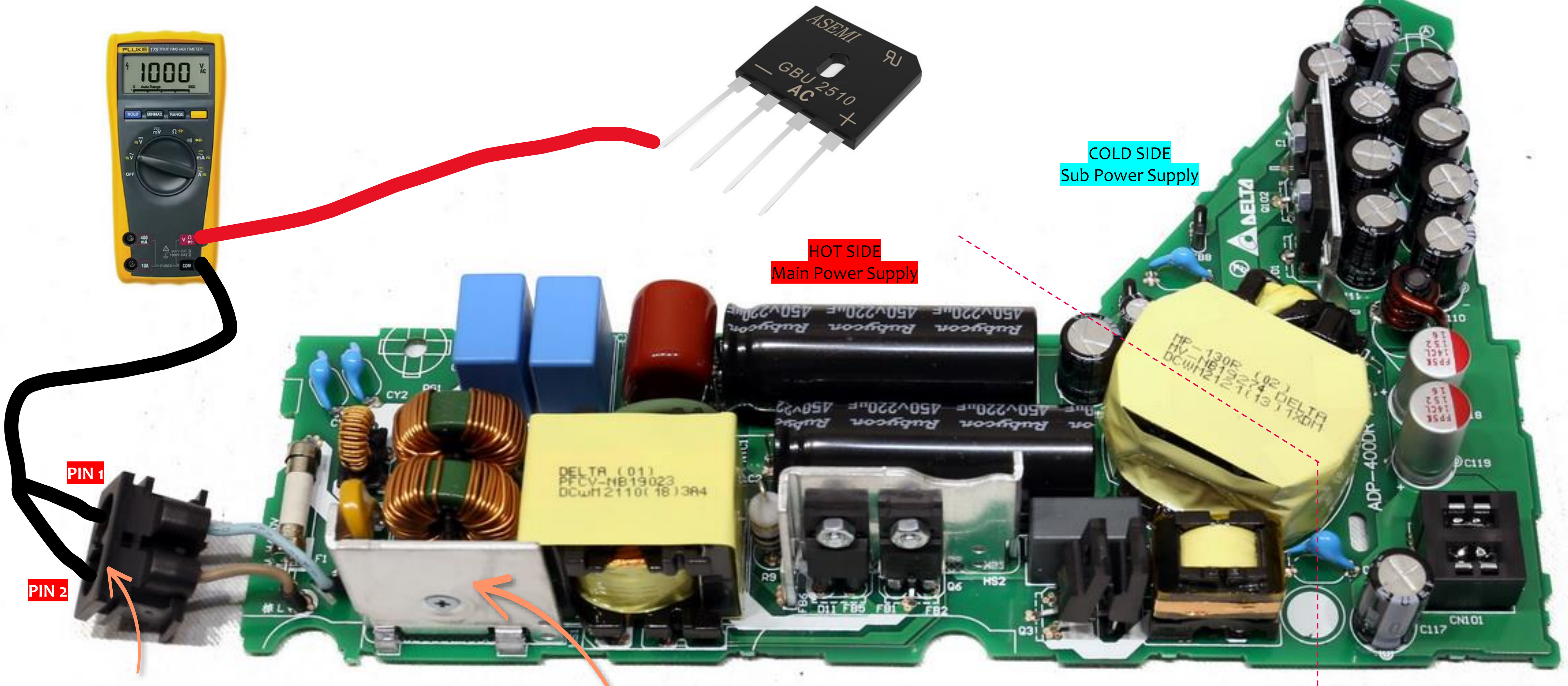
AC POWER ON!

Set multimeter in AC mode en measure the AC pins of the bridge rectifier

Multimeter in AC mode!

Voltage Testing – 1.1.2 AC input voltage

AC POWER OFF!



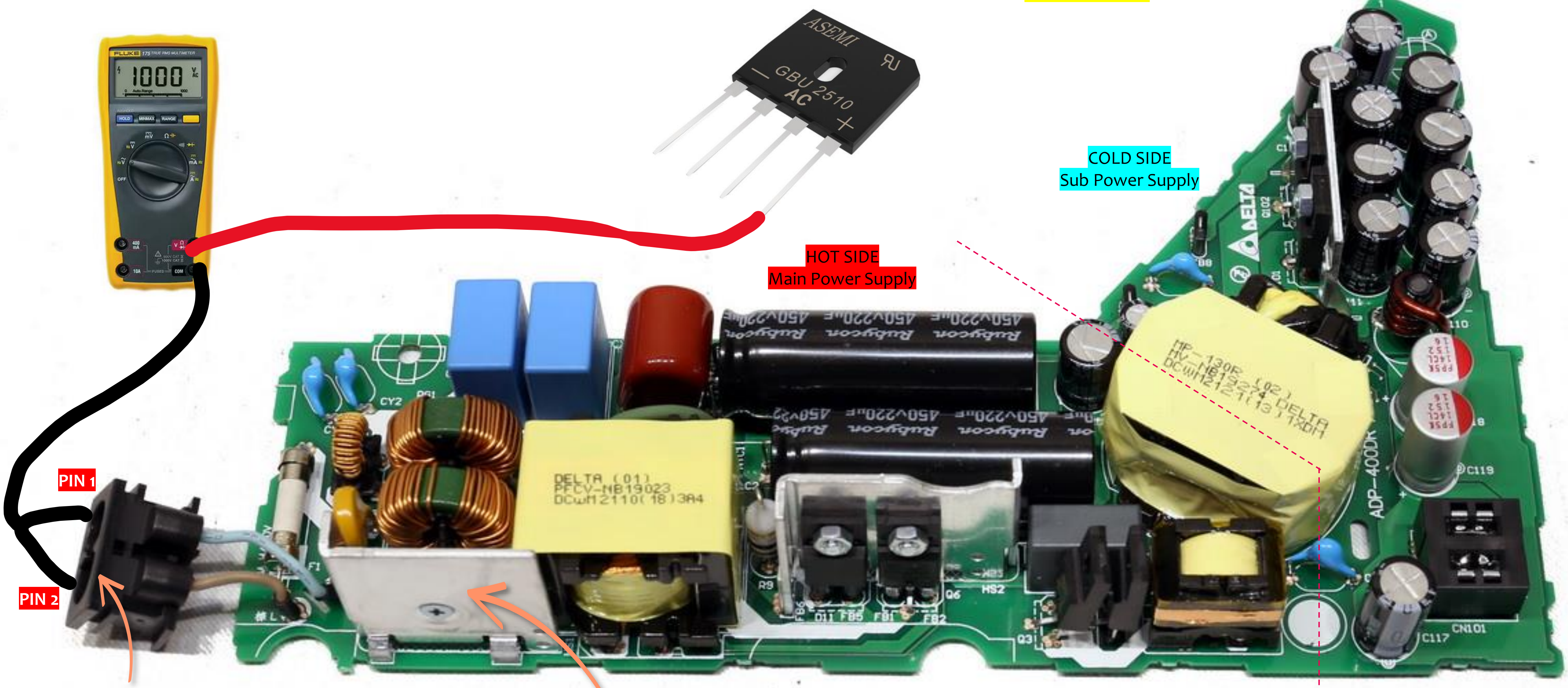
You will see the diode junction with voltage drop of 0.5V on pin 1 and pin 2. If yes circuit is good!

If not reverse the leads

Set multimeter in DIODE mode and measure the + and - pins of the bridge rectifier

Voltage Testing – 1.1.2 AC input voltage

AC POWER OFF!

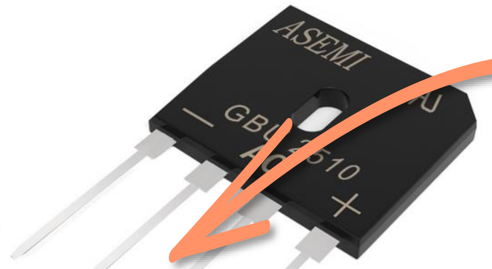


You will see the diode junction with voltage drop of 0.5V on pin 1 and pin 2. If yes circuit is good!

If not reverse the leads

Set multimeter in DIODE mode and measure the + and - pins of the bridge rectifier

Voltage Testing – 1.1 AC input voltage



Possibke faults

AC cable unplugged

AC cable broken

Main fuse open

Circuit traces open / dry joint

EMI coils may have open

Voltage Testing – 1.2 Main filter capacitor

If measured voltage is +- 390V then PFC circuit is working

PFC

Power factor correction (PFC) is a technique used in electrical systems to improve the power factor of an AC power supply. The power factor is a measure of how effectively electrical power is being utilized by a device or system. It represents the ratio of real power (measured in watts) to apparent power (measured in volt-amperes).

< 390V – PFC is not working correctly

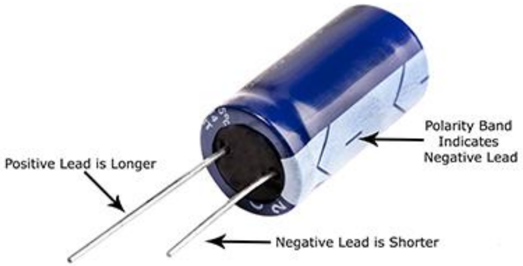
> 390V – PFC is not working correctly

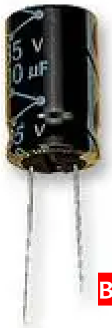
0V – Check circuit behind capacitor

Lower voltage then normal check health capacitor with ESR meter

+- 390V – PFC is ok when a load is connected to the 12V output of 8 – 10 amps

+- 380V – PFC is ok when NO load is connected to the 12V output





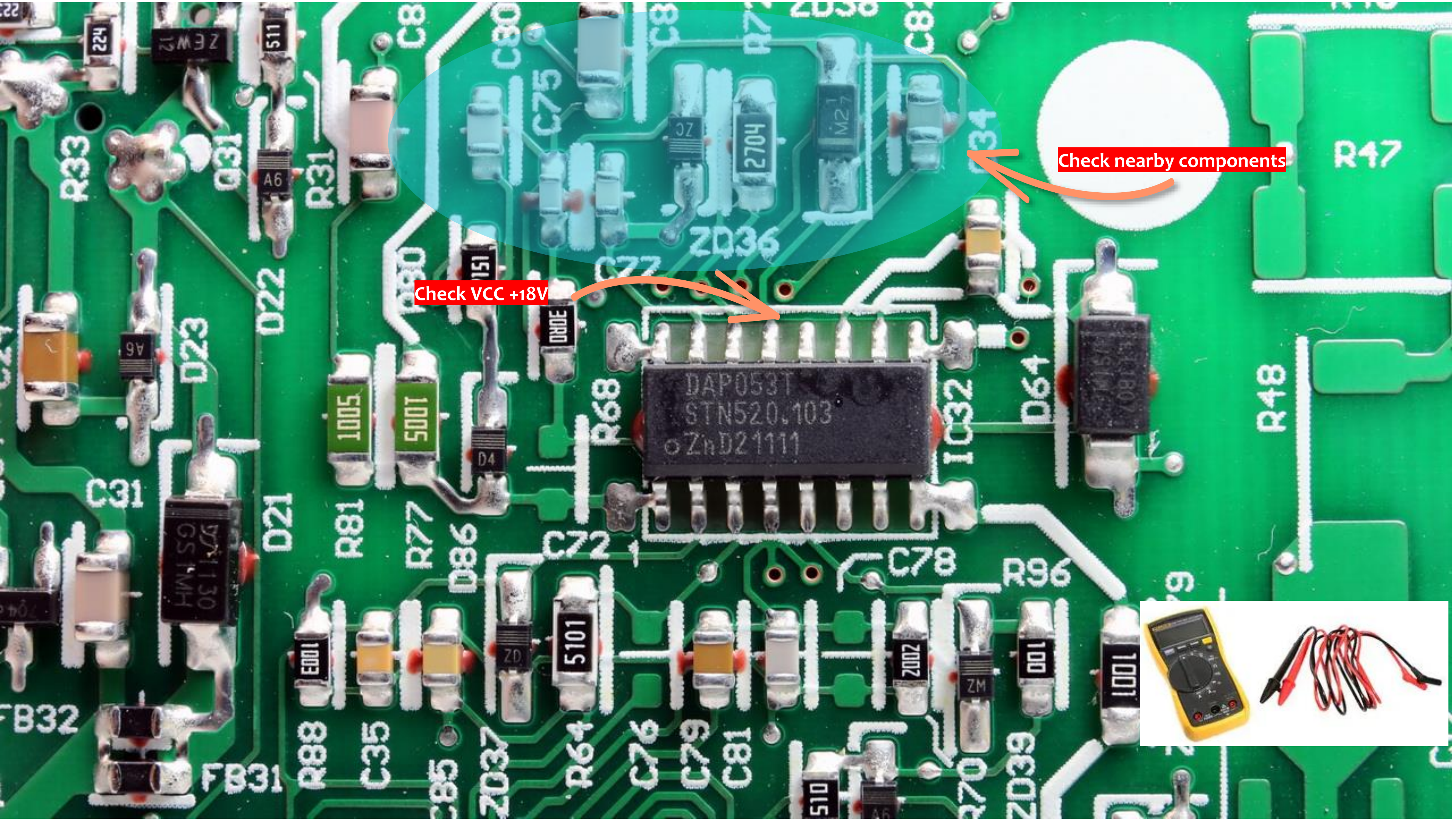
Black probe

Red probe

Multimeter in DC mode!
Don't let the probes touch other pins!

Testing done?
Power off! - discharge capacitor

Voltage Testing – 1.3 Supply voltage VCC – PWM IC



VCC - PIN

Identify the VCC Pin of the PWM / Power IC: Look at the datasheet or pinout diagram of the specific PWM power IC you are using. The VCC pin is usually marked clearly in the documentation.

For PS5 it is **+18V.**

VCC is needed to power the PWM / Power IC on to generate the voltages.

Correct voltage is present. We can assume now the filter capacitor; startup resistor and bridge rectifier are ok.

0V - and main filter capacitor has the correct voltage then the startup resistor could be shorted pulling the VCC down to 0V.

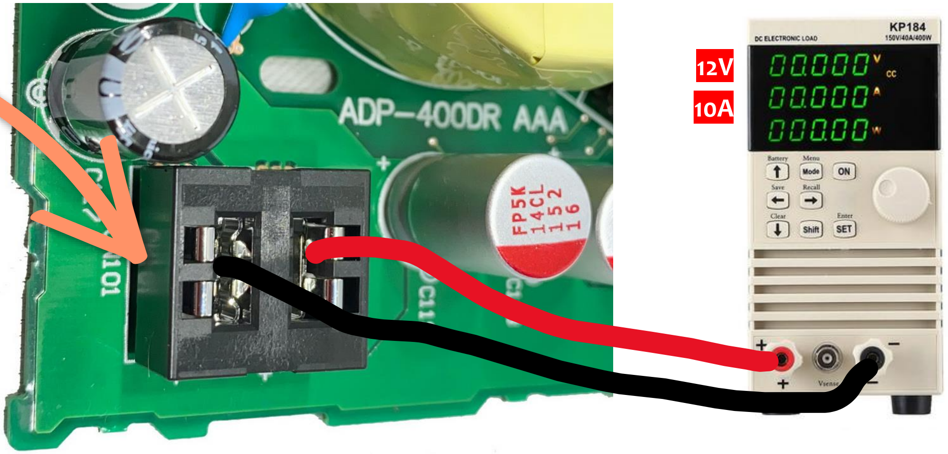
Voltage lower then aspected and main filter capacitor has the correct voltage then the resistance of startup resistor could be high ohm. Or components nearby are leaky or faulty

AC POWER ON!

Voltage Testing – 1.4 Secondary Output Voltages

AC POWER ON!

Check output voltage



To test output voltage, you need to connect an electronic load. For PS5 it is +12V and 8 Amp of current

12V

10A

Testing output voltages on the cold side you need to know what voltages are generated.

For PS5 it is 12V / 10 amp

Some SMPS have multiple outputs with different voltages.

Output voltage is OK, no action needed.

0V - SMPS is faulty or some short is preventing the SMPS from starting

Low output voltages then specified. Problem could be on the hot or cold side. Current sense resistor could have a different value or one of the secondary filter capacitor has a high ESR.

Higher output voltage. If it is too high the SMPS will shutdown. When slightly high the problem could be in the feedback circuit area, a resistor or optoislator turned into high ohm.

Output is power cycling. Max voltage to zero voltage and then it repeats. Cause can be in hot or cold side.

SMPS shutdown. You will see that the ouput voltage goes up and then down. When you switch the SMPS off and on the same thing happens.

Voltage Testing – 2.1 Use oscilloscoop to test waveforms

DO NOT TOUCH THE ROUND METAL CONNECTORS!

Use the Isolator transformer
When you are measuring the hot side with a oscilloscope!

HOT SIDE
Main Power Supply

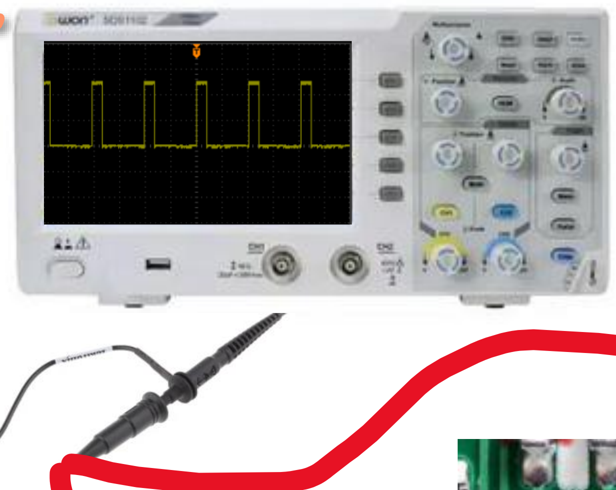
Test PWM signal of power IC on the HOT side

COLD SIDE
Sub Power Supply

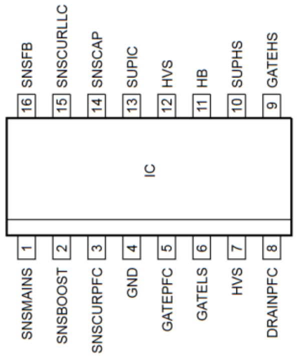
Test waveform on the DC output of the cold side at the output of the diodes.

In PS5 this would be testing the MOSFET's and IC that drives the mosfets and the output line.

Voltage Testing – 2.1 Use oscilloscoop to test PWM waveforms

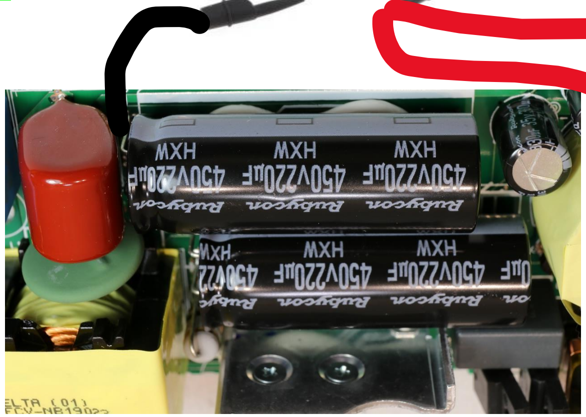


Set oscilloscope
in AC mode
Set the time and voltage



HOT SIDE
Main Power Supply

Test PWM signal on the gate output of high and low side mosfet



Gate high side mosfet

Gate low side mosfet

Good output signal means: IC is getting power, components near the IC are working. Troubleshoot the cold side.

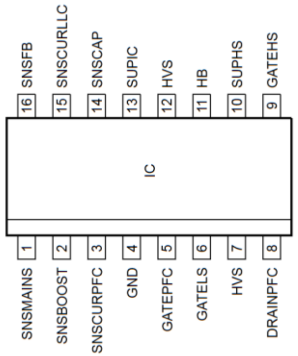
No output signal but the IC is getting correct power then: IC could be faulty , Mosfet is shorted, Primary winding of main transformer is shorted or a short on the cold side.

Voltage Testing – 2.2 Use oscilloscoop to test drain of mosfet

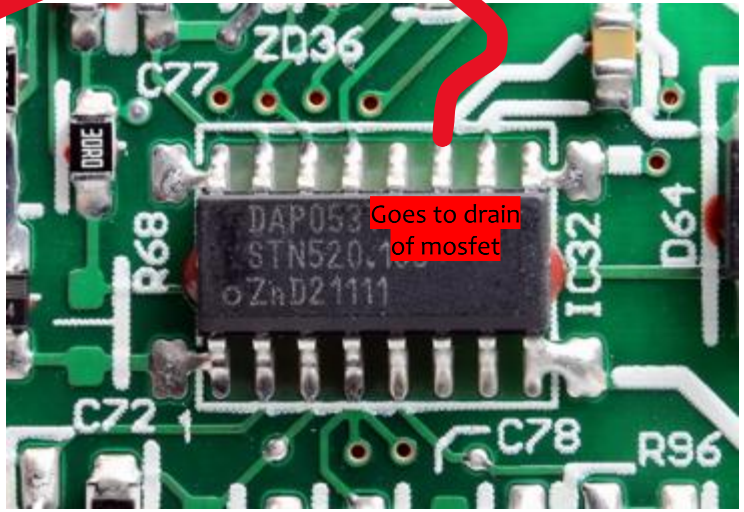
HOT SIDE
Main Power Supply

Test drain of mosfet

Set oscilloscope
in AC mode
Set the time and voltage



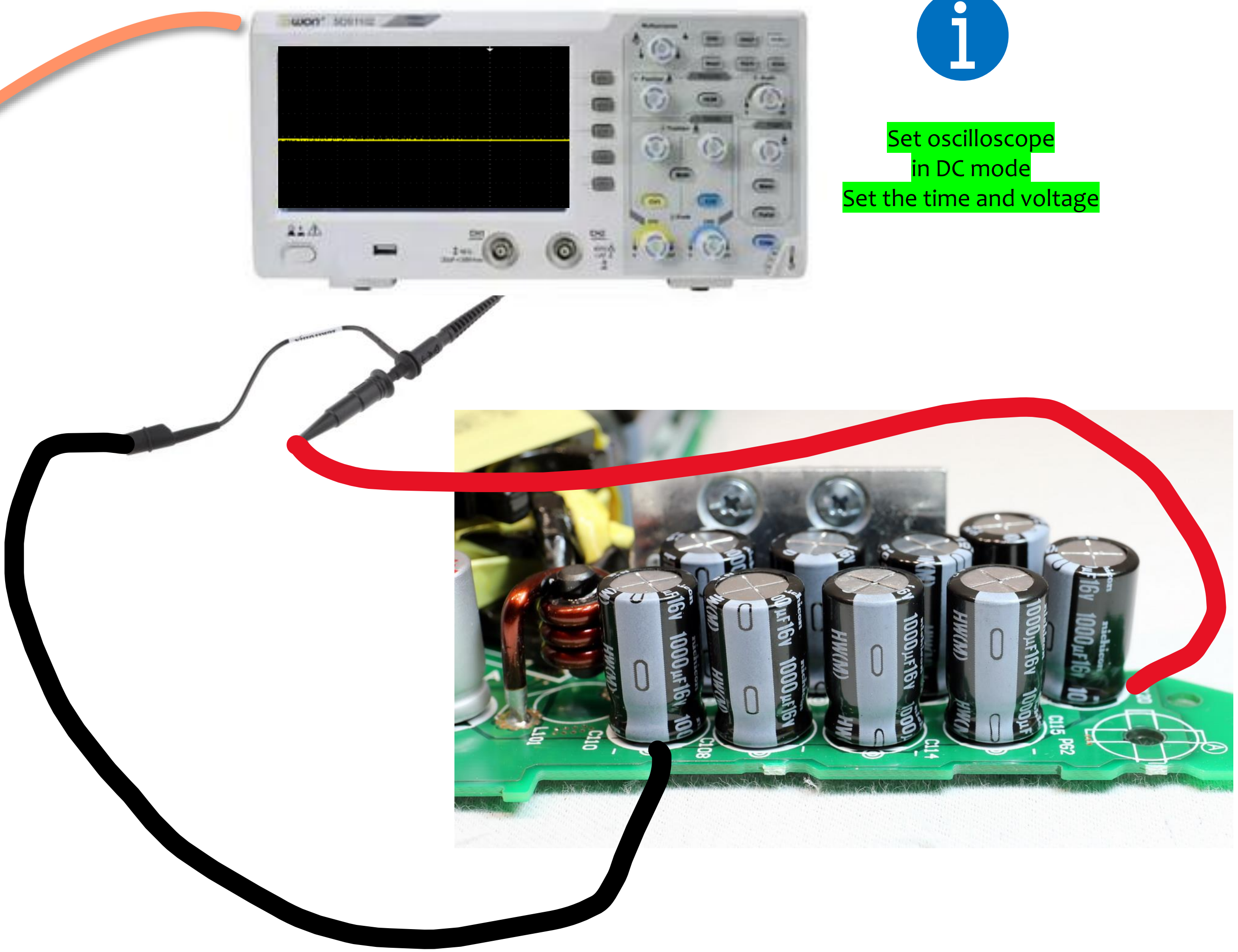
See the signal of the drain pin.



Goes to drain
of mosfet

Voltage Testing – 2.2 Use oscilloscoop to test DC output voltage

COLD SIDE
Sub Power Supply

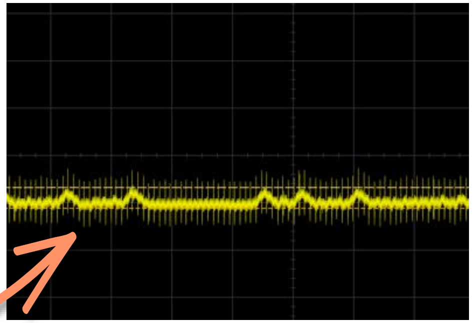


Set oscilloscope
in DC mode
Set the time and voltage

The DC output voltage generated should be a clean DC voltage. On the scope the waveform should be a straight horizontal line without distortion and ripples.

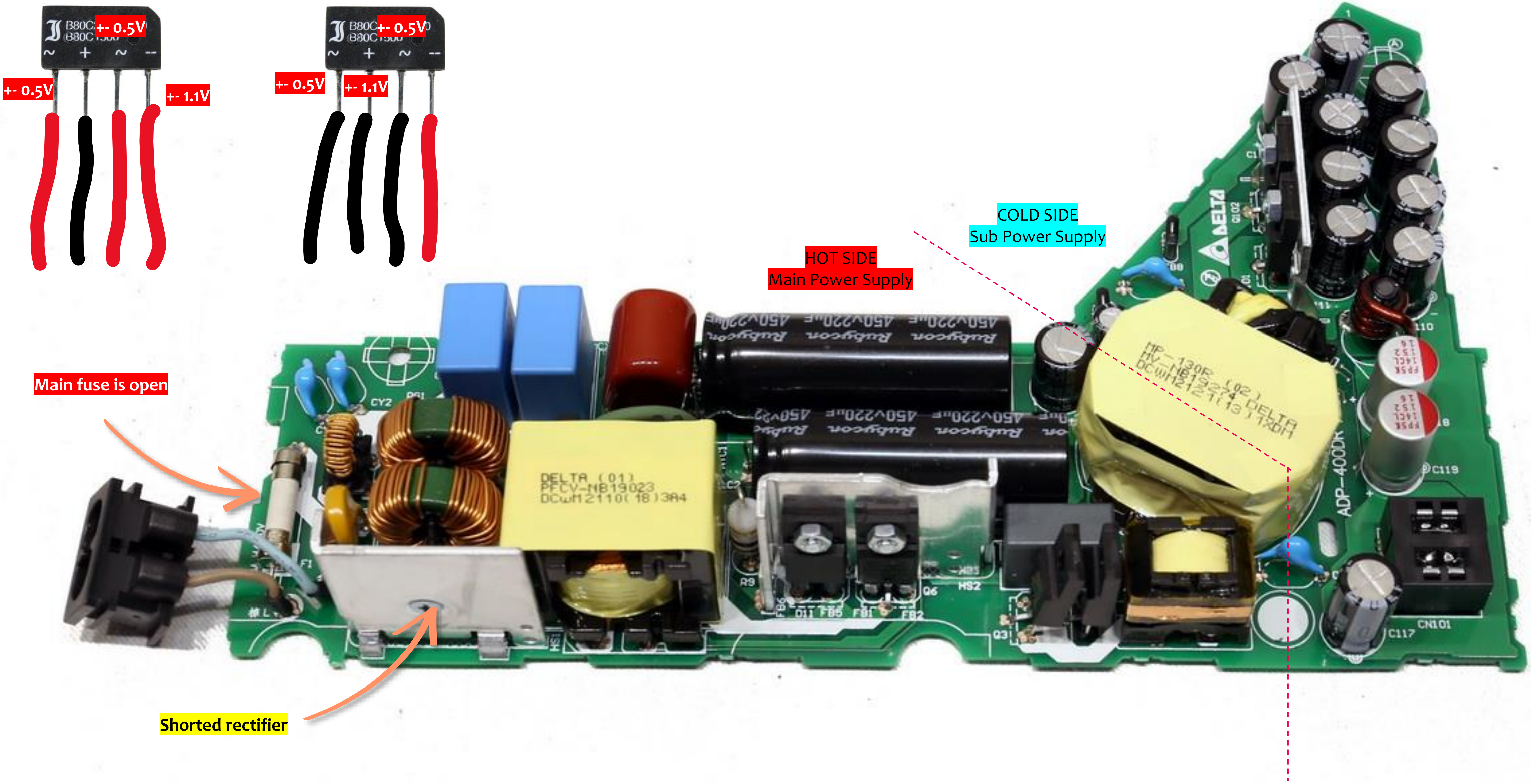
When the DC output voltage is clean you can assume the main and secondary filter capacitors are working ok.

When the DC output voltage has ripple you need to check all the filter capacitors with a ESR meter.



Ripple in DC voltage,
this is not good

Troubleshooting – Bridge rectifier



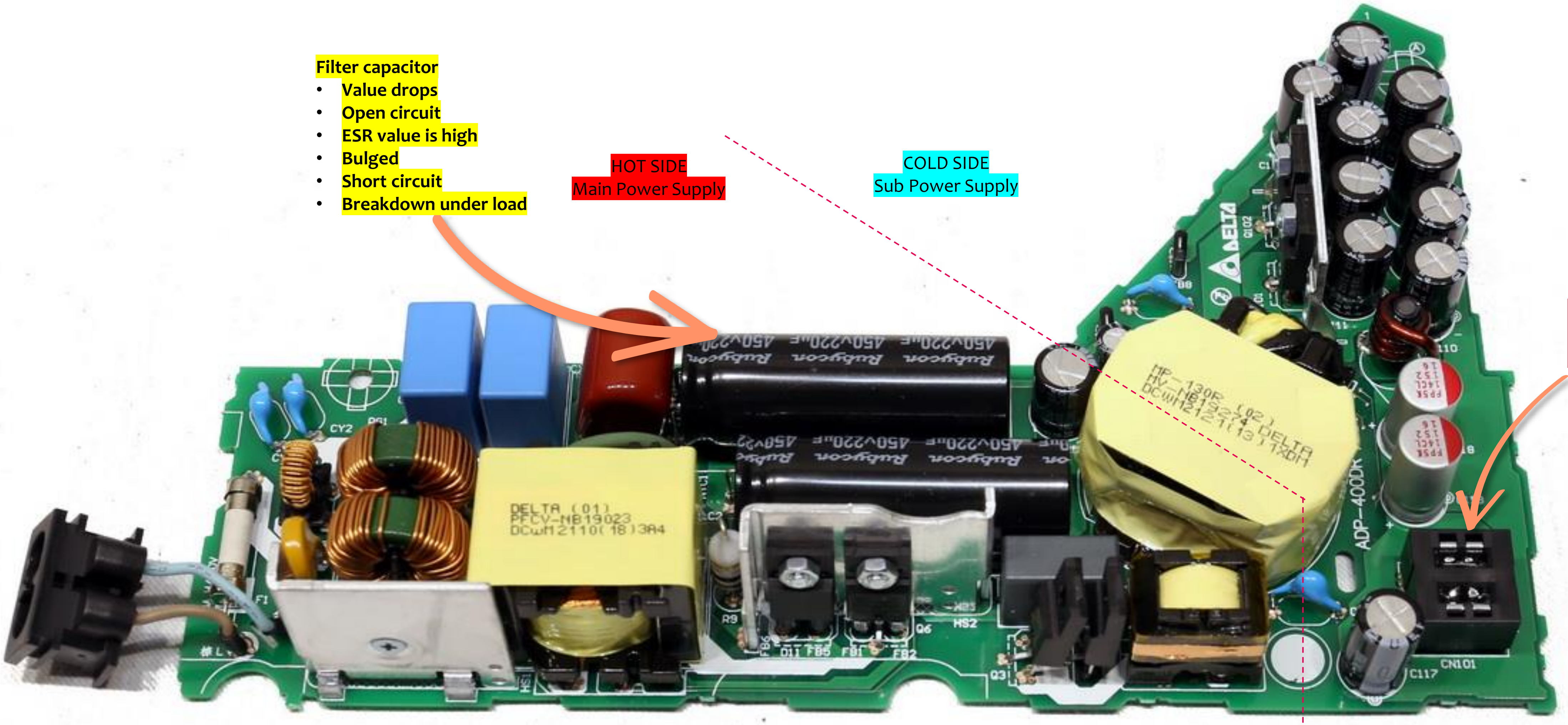
Troubleshooting – Main filtering capacitor

Filter capacitor

- Value drops
- Open circuit
- ESR value is high
- Bulged
- Short circuit
- Breakdown under load

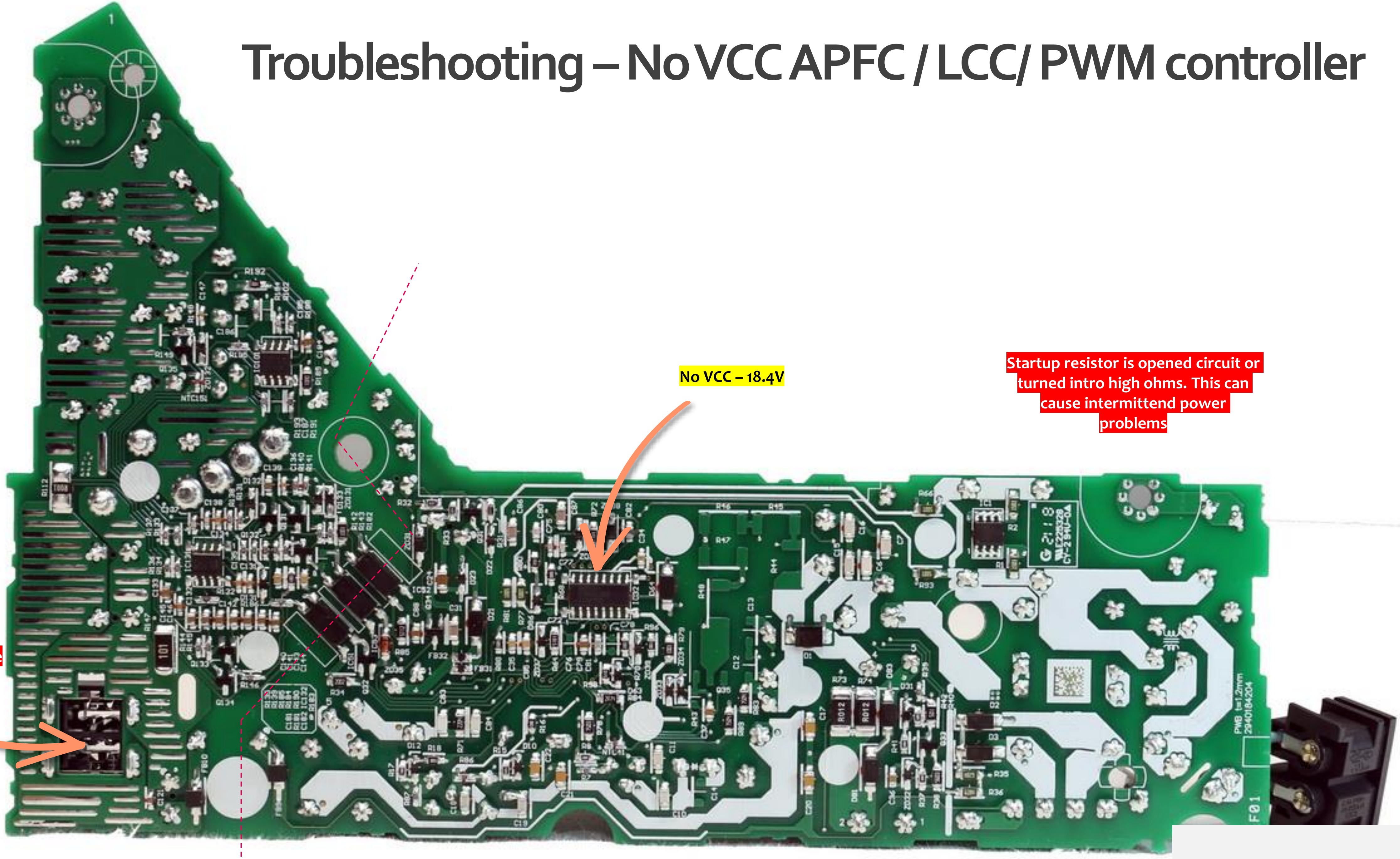
HOT SIDE
Main Power Supply

COLD SIDE
Sub Power Supply



- No voltage
- Voltage blink
- Unstable voltage

Troubleshooting – No VCC APFC / LCC/ PWM controller

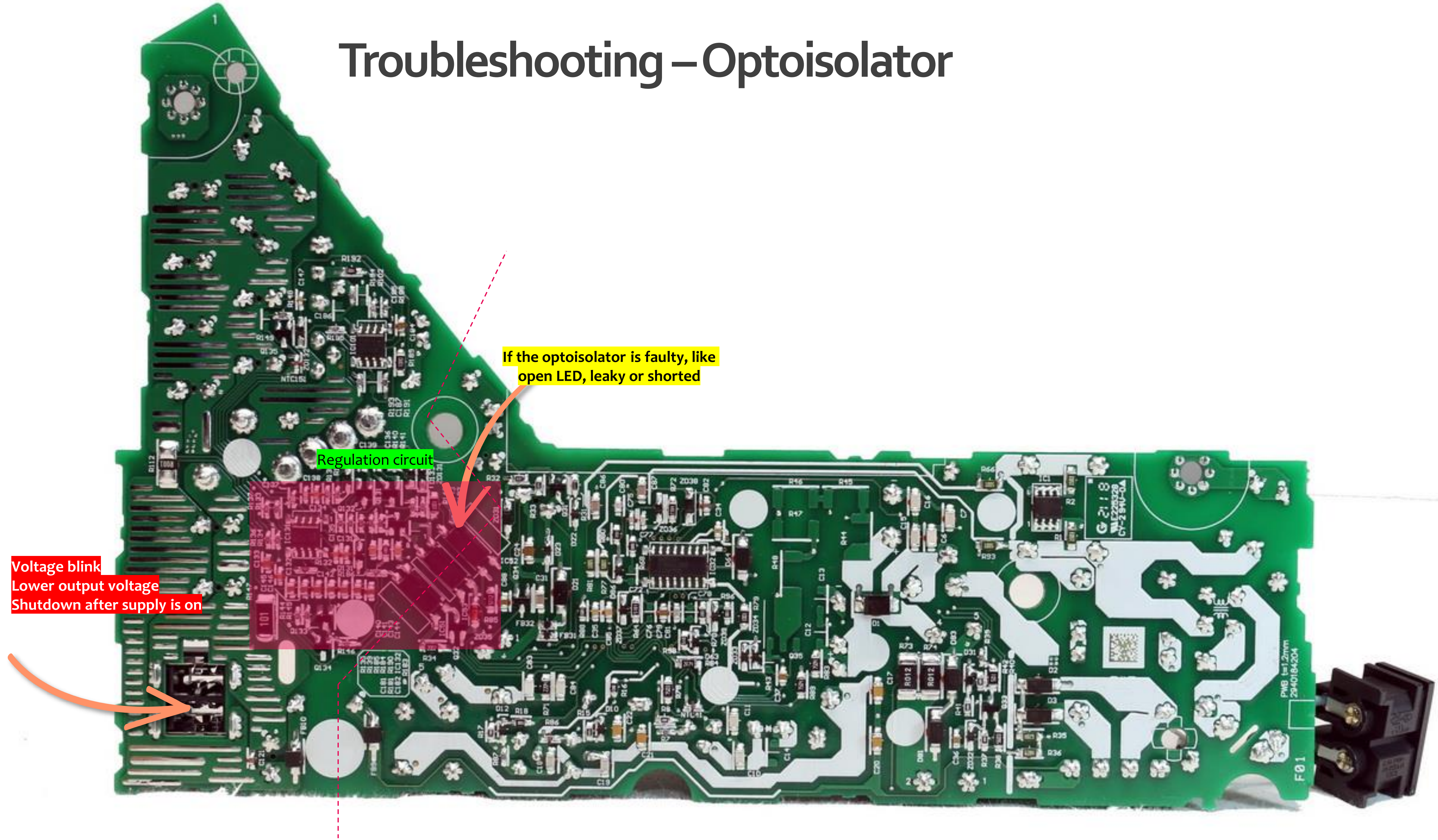


No VCC – 18.4V

Startup resistor is opened circuit or turned intro high ohms. This can cause intermittend power problems

No voltage

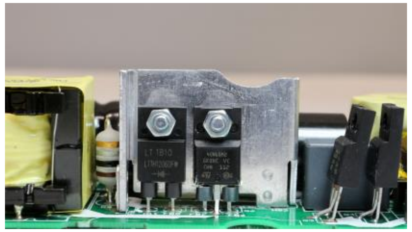
Troubleshooting – Optoisolator



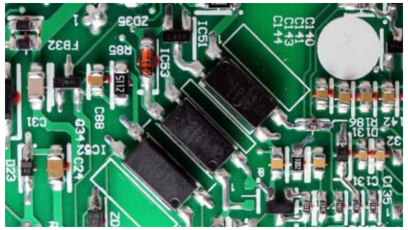
APFC / PWM controller IC

DAP053T

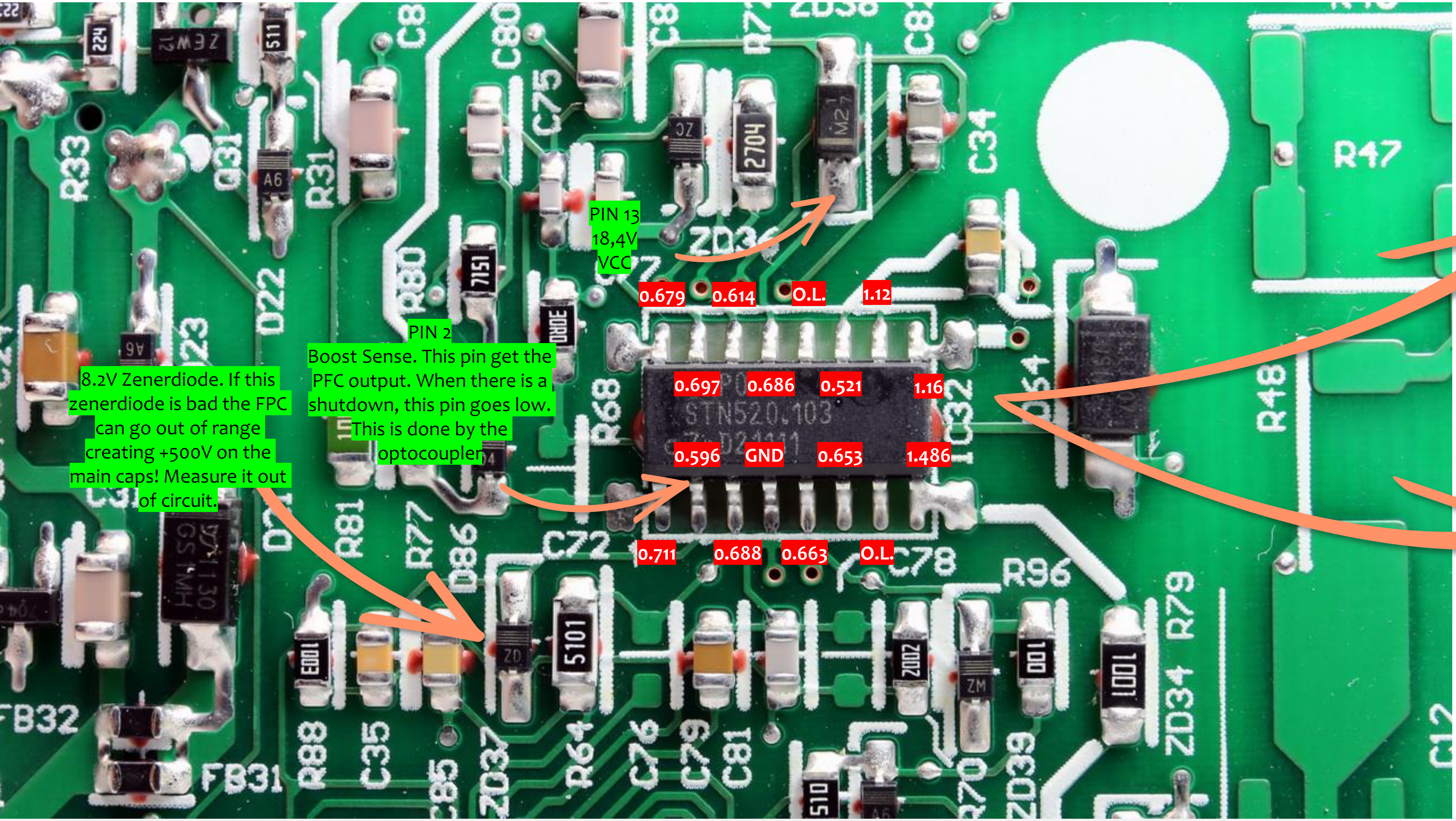
Measure IC for shorts and measure each pin in diode mode using the multimeter



APFC Boost Diode
APFC MOSFET
LLC High and Low side mosfet



Powergood
Feedback
Shutdown



= diode measurements

APFC / PWM controller IC

DAP053T = TEA2016AAT

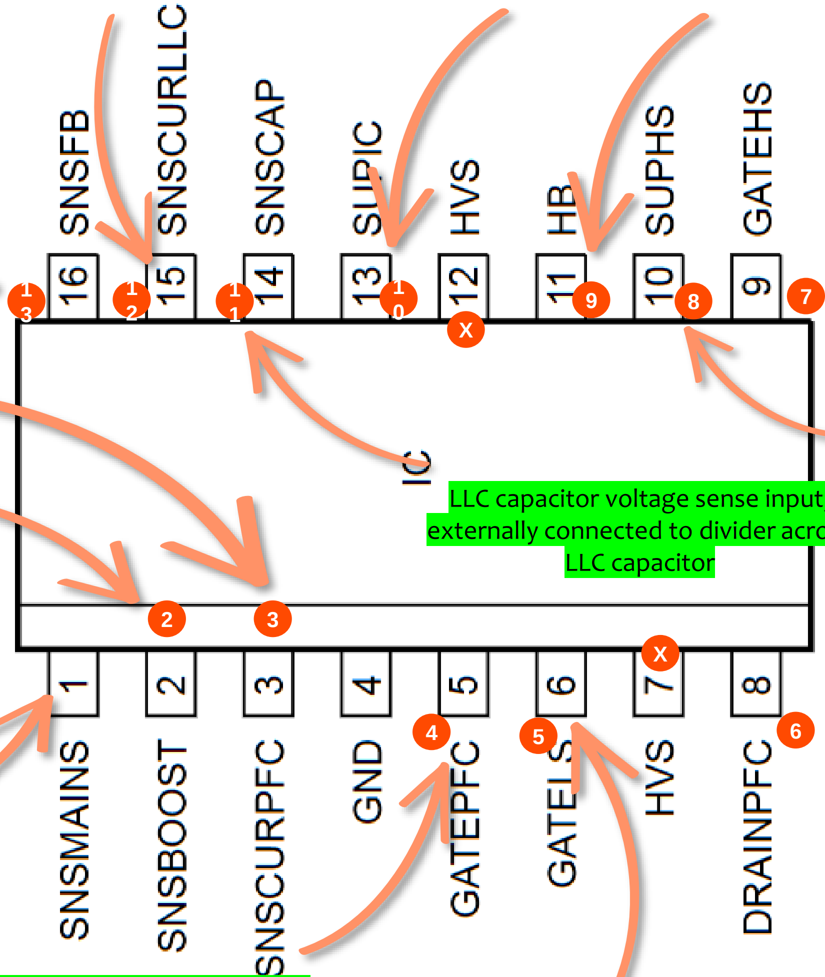
LLC current sense input; externally connected to the resonant current sense resistor

VCC – 18.4V Input supply external DC supply

Input for half-bridge slope detection; externally connected to half-bridge node HB between the LLC MOSFETs

output voltage regulation feedback sense input; externally connected to an optocoupler. Output for power good function

High-side MOSFET gate driver output and supply for bootstrap Capacitor (LLC)



PFC current is measured via an external sense resistor connected to this pin. (PFC)

high-side driver supply input

Sense input for boost voltage (PFC)

LLC capacitor voltage sense input; externally connected to divider across LLC capacitor

The AC input voltage is measured, and the information is fed to an internal mains compensation circuit (PFC)

Start-up source also used for X- capacitor discharge, valley detection, and PFC OVP detection; connected to (PFC) drain voltage)

PFC MOSFET gate driver output (PFC)

Low-side MOSFET gate driver output and supply for bootstrap Capacitor (LLC)

APFC / PWM controller IC

DAP053T = TEA2016AAT

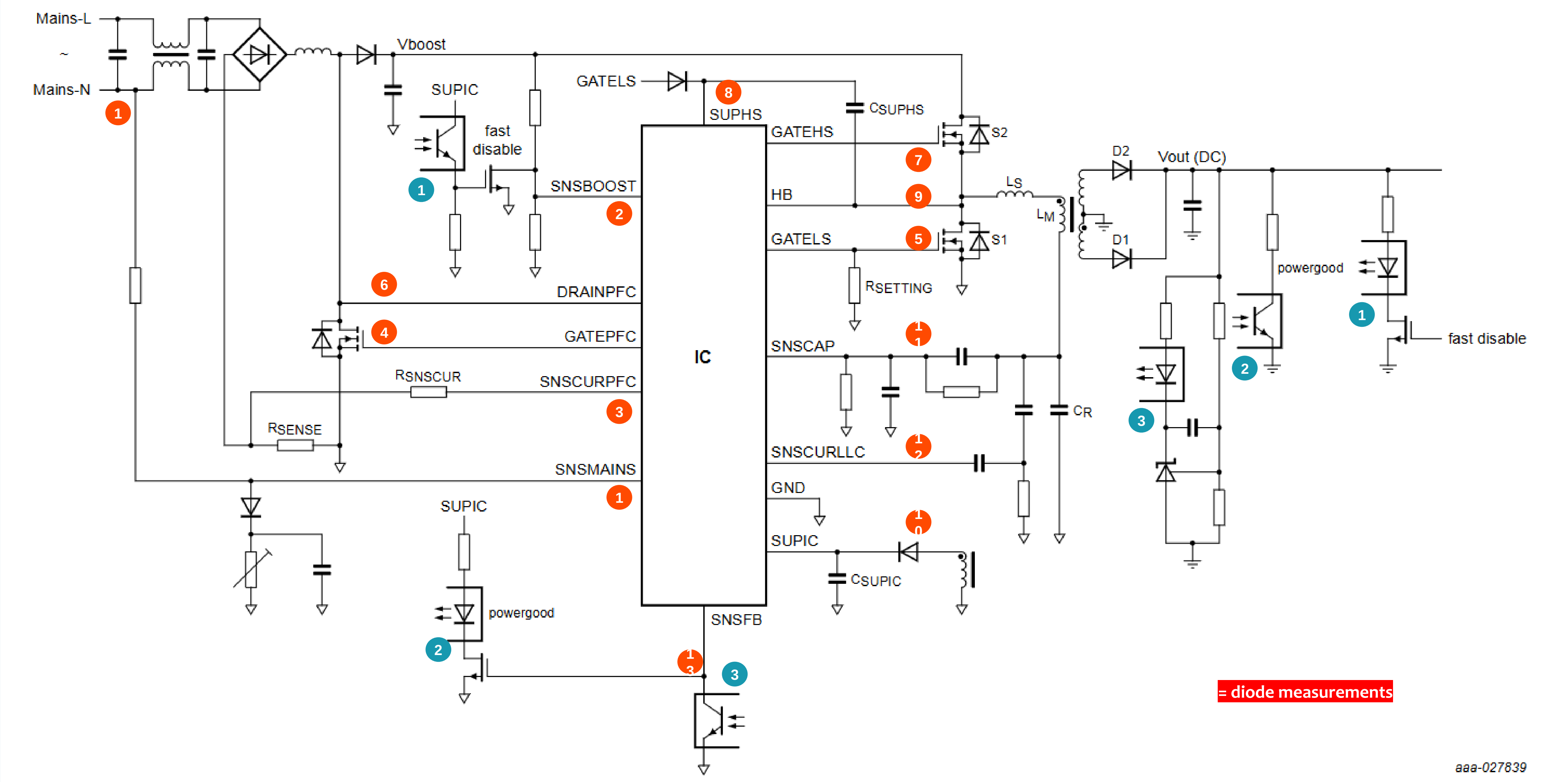
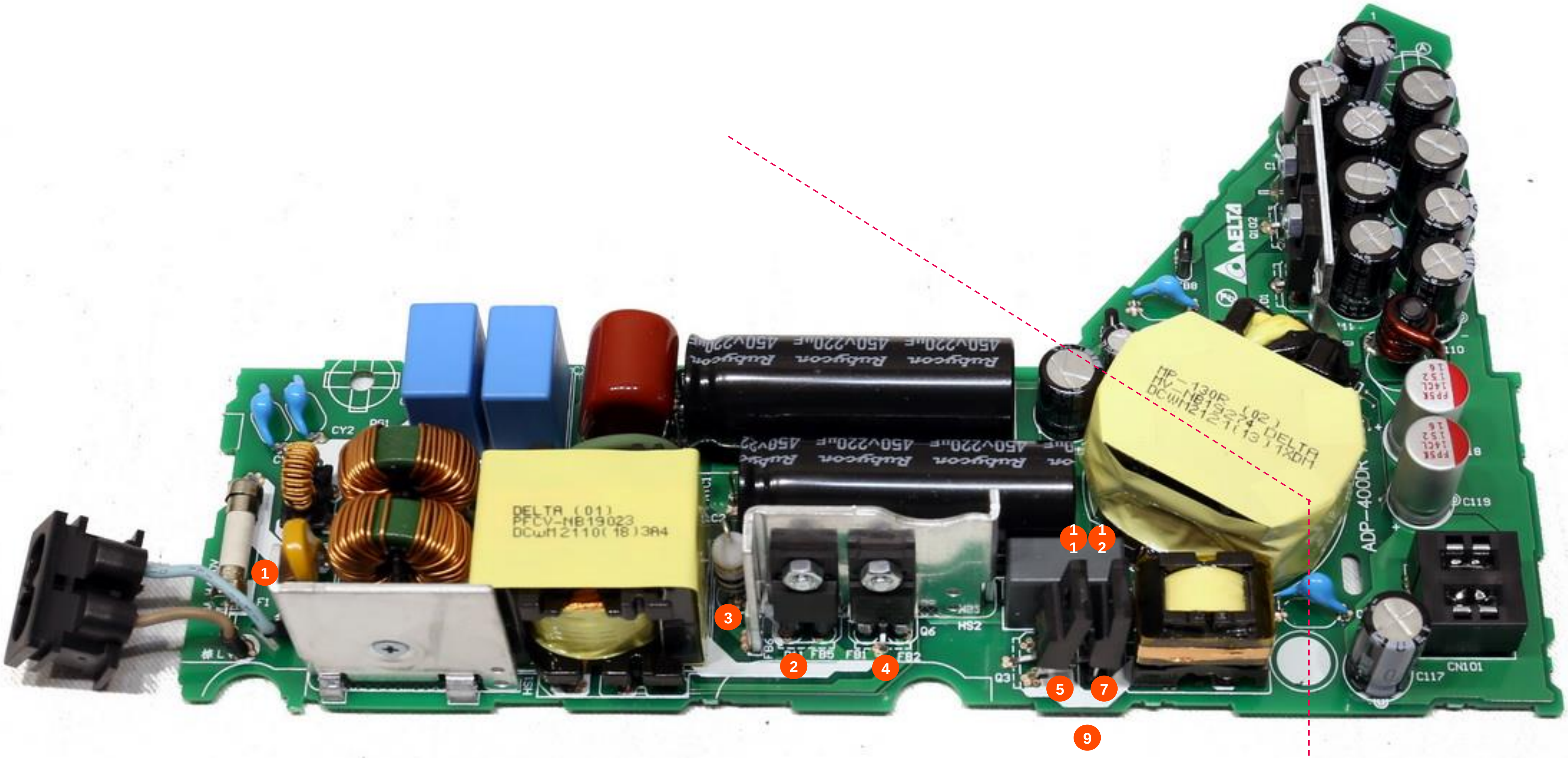
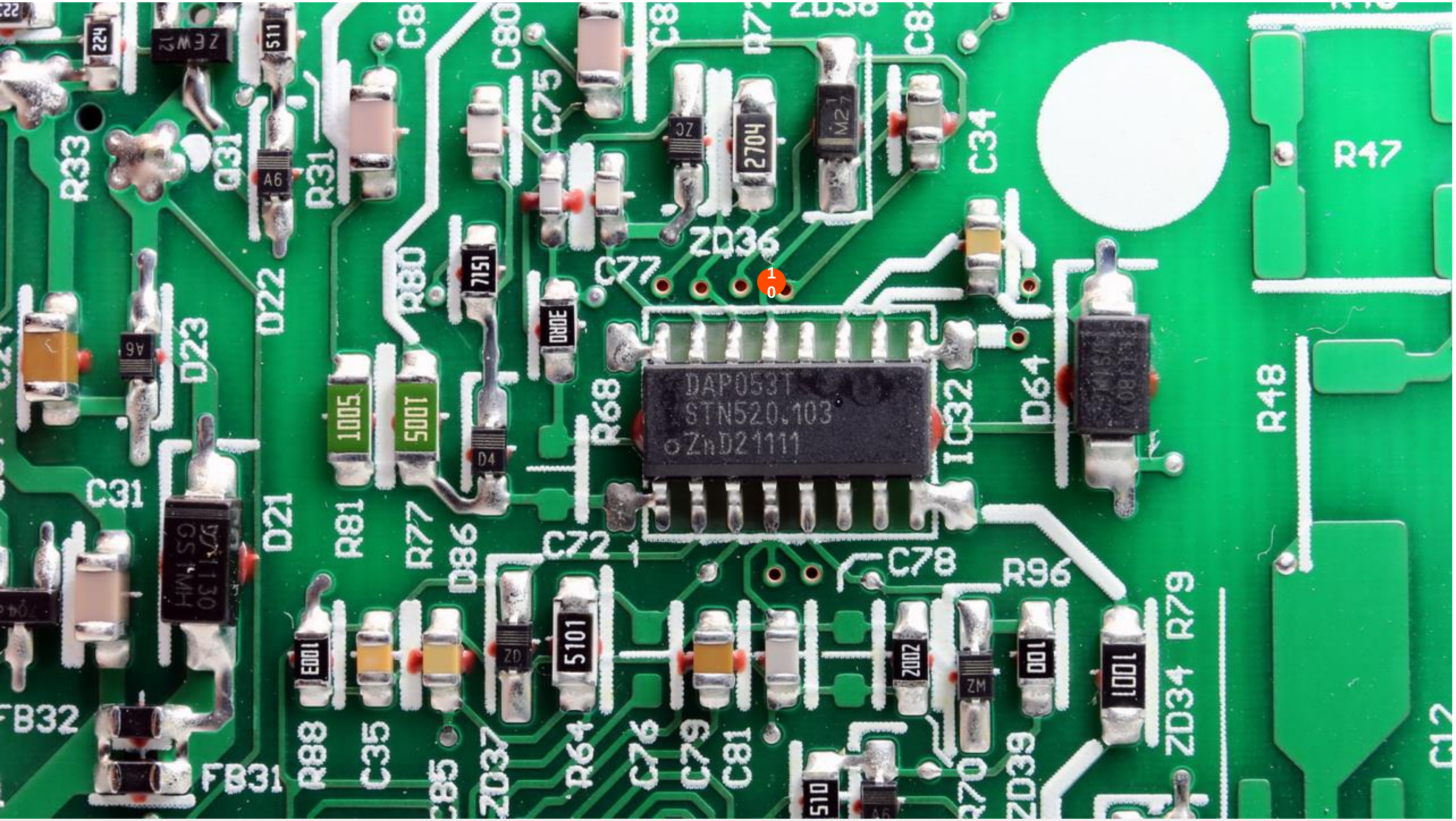


Figure 25. TEA2016AAT application diagram

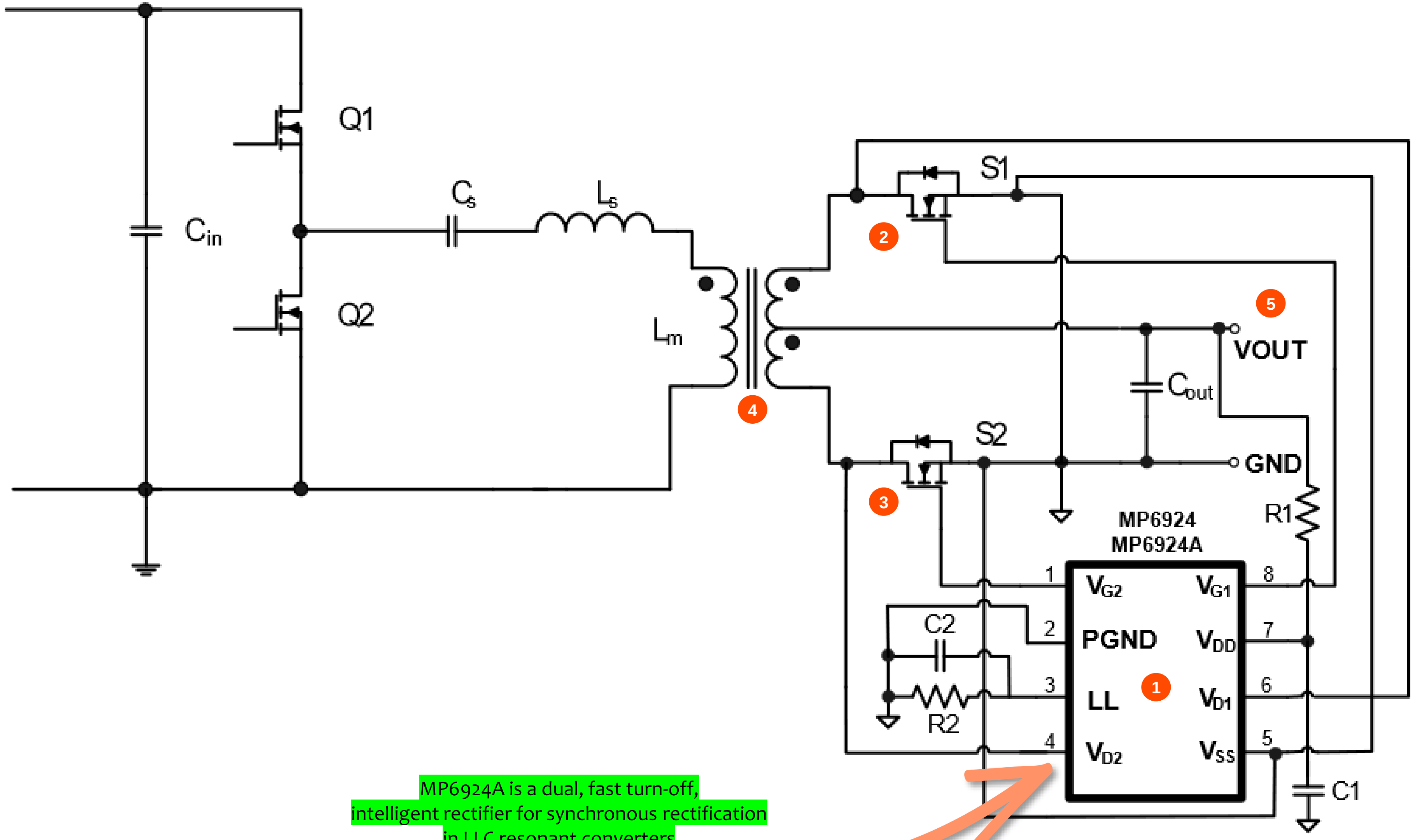
Front



APFC / PWM controller IC



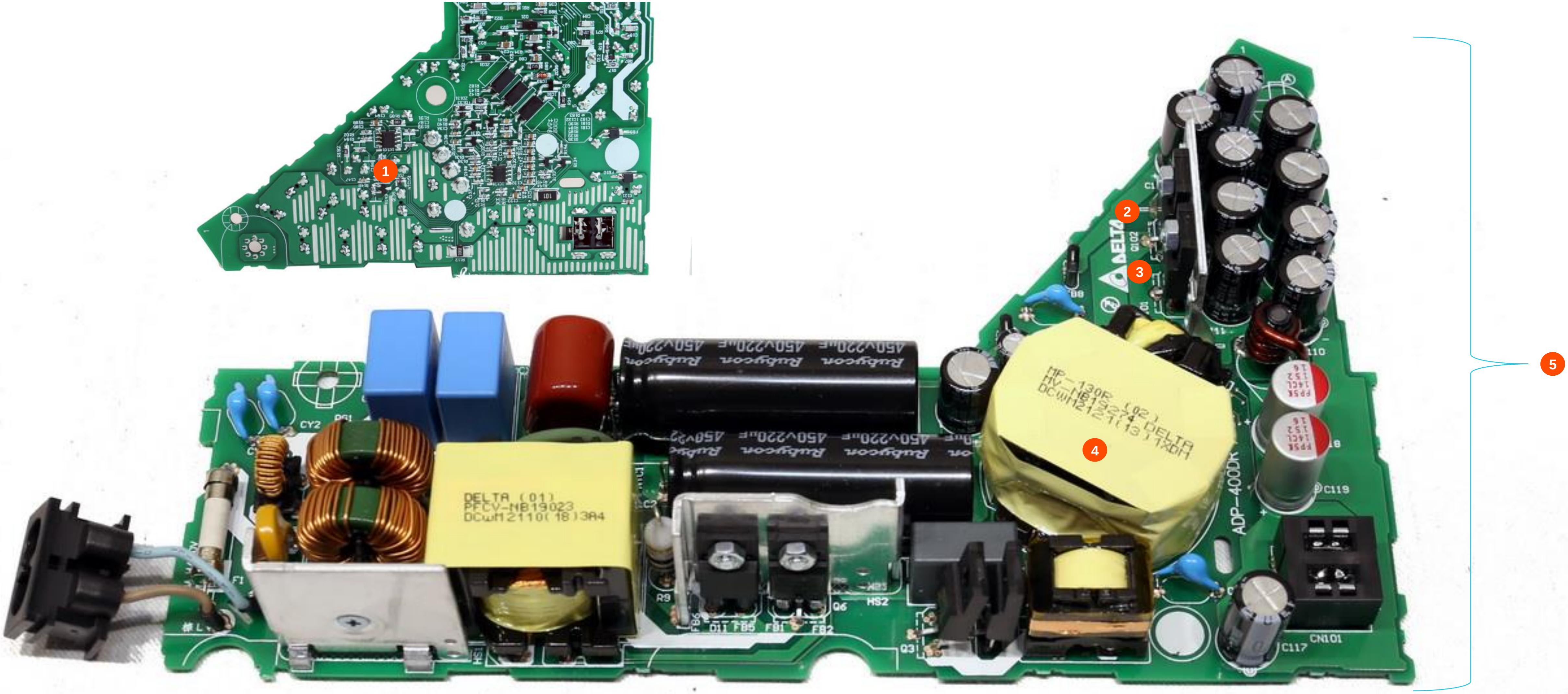
Synchronous rectification



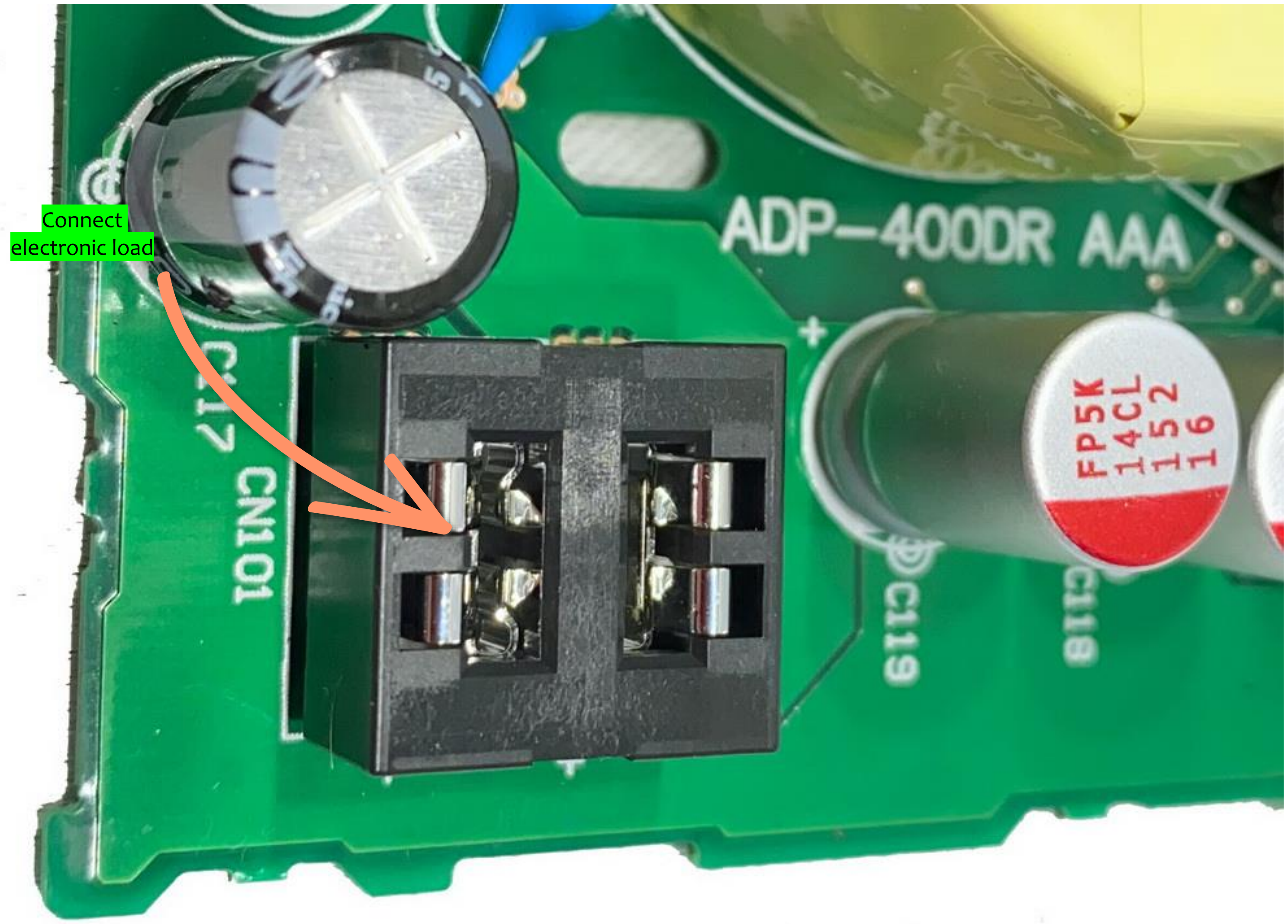
MP6924A is a dual, fast turn-off, intelligent rectifier for synchronous rectification in LLC resonant converters

Synchronous rectification

MP6924A



Power supply failing under load



Conditions

The PS5 power supply can deliver more then 30 Amps and 12V.

If the electronic load is connected set the voltage to 12V and then increase load current in intervals of 1 Amp.

The voltage of 12V must not drop when you increase the load current until 10 amps.

When to voltage is dropping below 12V when you increase the current the power supply is failing under load.

Measure main filter capacitor under load and no load.

Witch load connected: 390V
No load connected: 385V

Measure APFC / PWM controller IC in diode mode for deviations in value.

To eliminate that the shutdown circuit is causing the problem disable the transistor that is connected to pin 2. if the main capacitor is still too low then replace the APFC / PWM IC.

Electronic load results

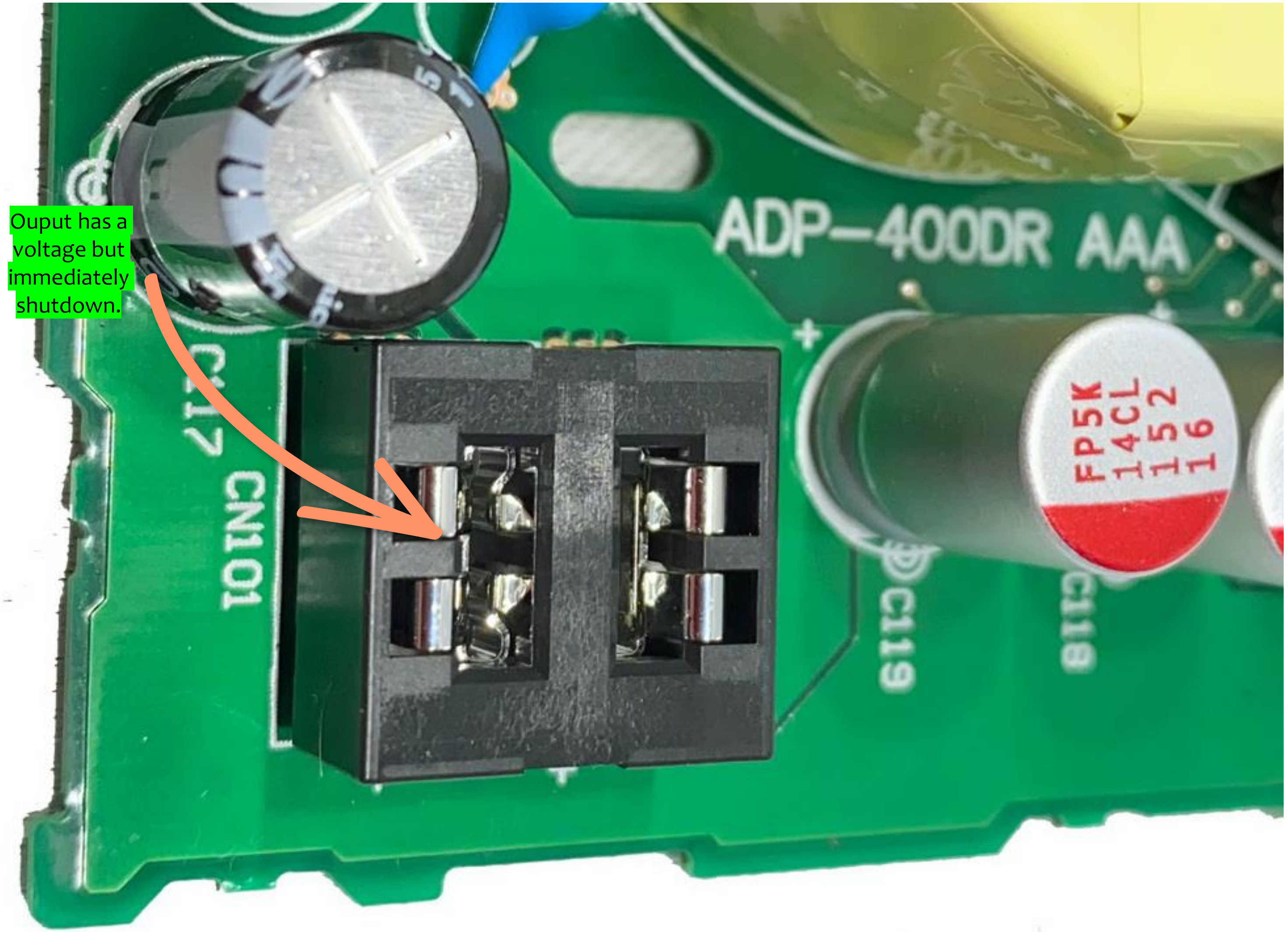
Working APFC

Load voltage (v)	Load current (A)	Main filter capacitor (V)
+12V	4	+- 390
+12V	5	+- 390
+11.9V	6	+- 390
+11.9V	7	+- 390
+11.9V	8	+- 390
+11.8V	9	+- 390
+11.8V	10	+- 390

Not correctly working APFC

Load voltage	Load current (A)	Main filter capacitor (V)
+12V	4	+- 390
+12V	5	+- 390
+11.8V	6	+- 385
+11.7V	7	+- 360
+11.7V	8	+- 360
+11.6V	9	+- 350
+11.6V	10	+- 330

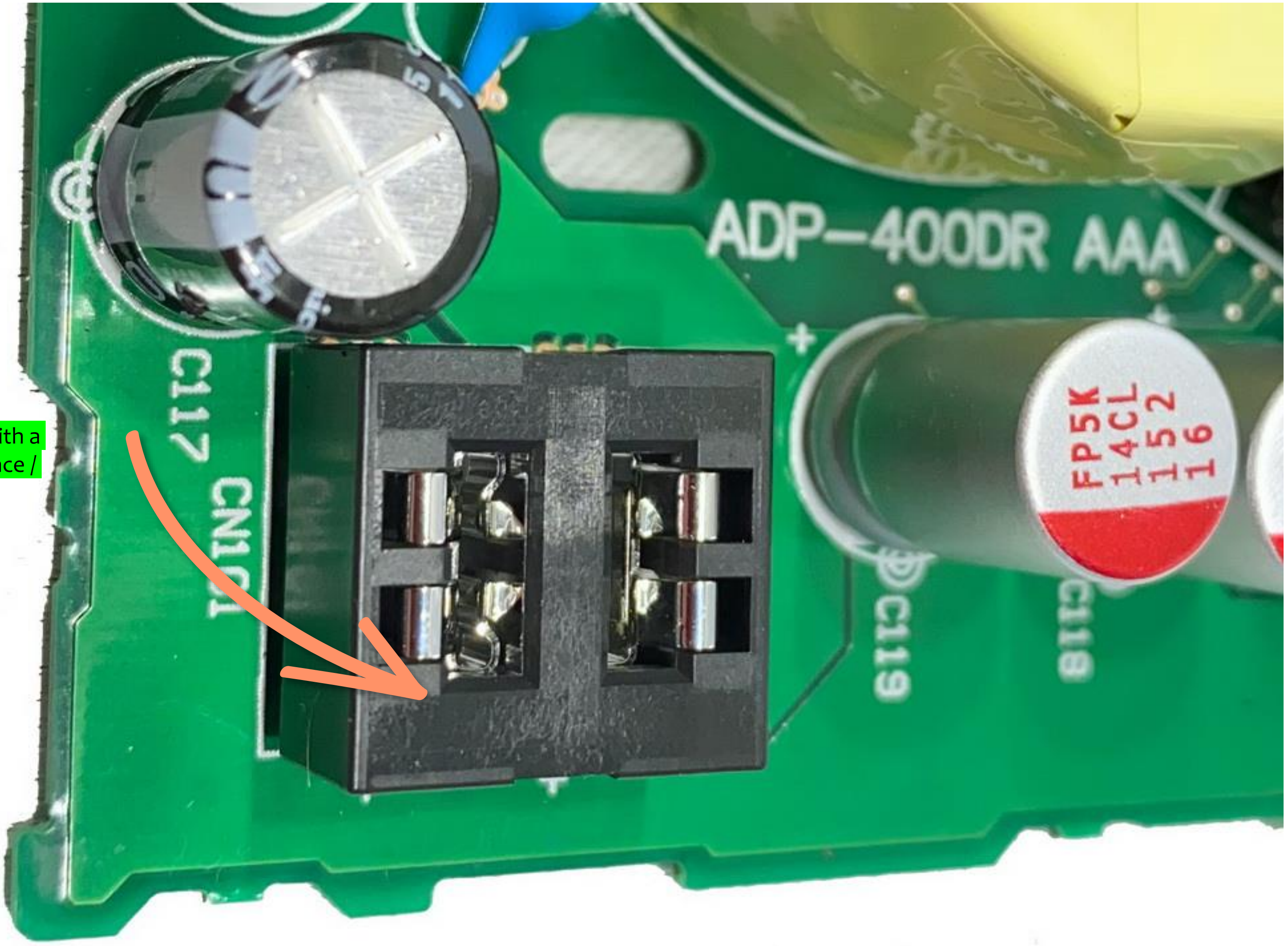
Fast shutdown on output



Shutdown symptoms

If you see a voltage being created on the output but then it goes to 0. This means that the power supply wants to generate the voltage, but something is triggering a shutdown.

Measure output for shorts

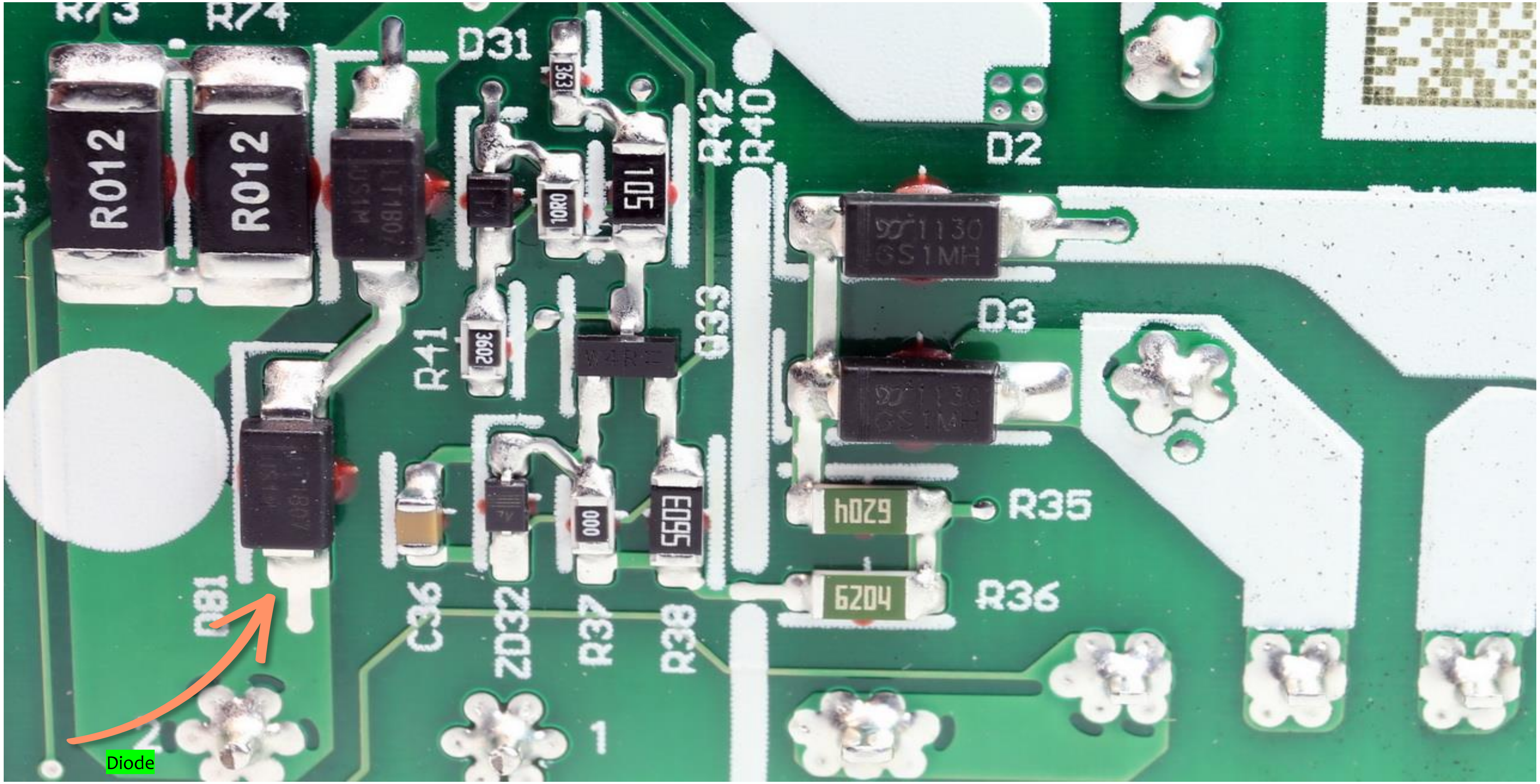


Measure the output with a multimeter in resistance / diode mode

If there is a short on the cold / secondary side. Check all capacitors and rectifier diodes

+12V Output shorted – possible causes

Check for shorts and search for burn marks



Fusible resistor

Fusable resistor

A fusible resistor, also known as a fuse resistor or a safety resistor, is a type of resistor that is designed to act as a fuse in case of excessive current or power dissipation. It combines the functions of a resistor and a fuse into a single component.

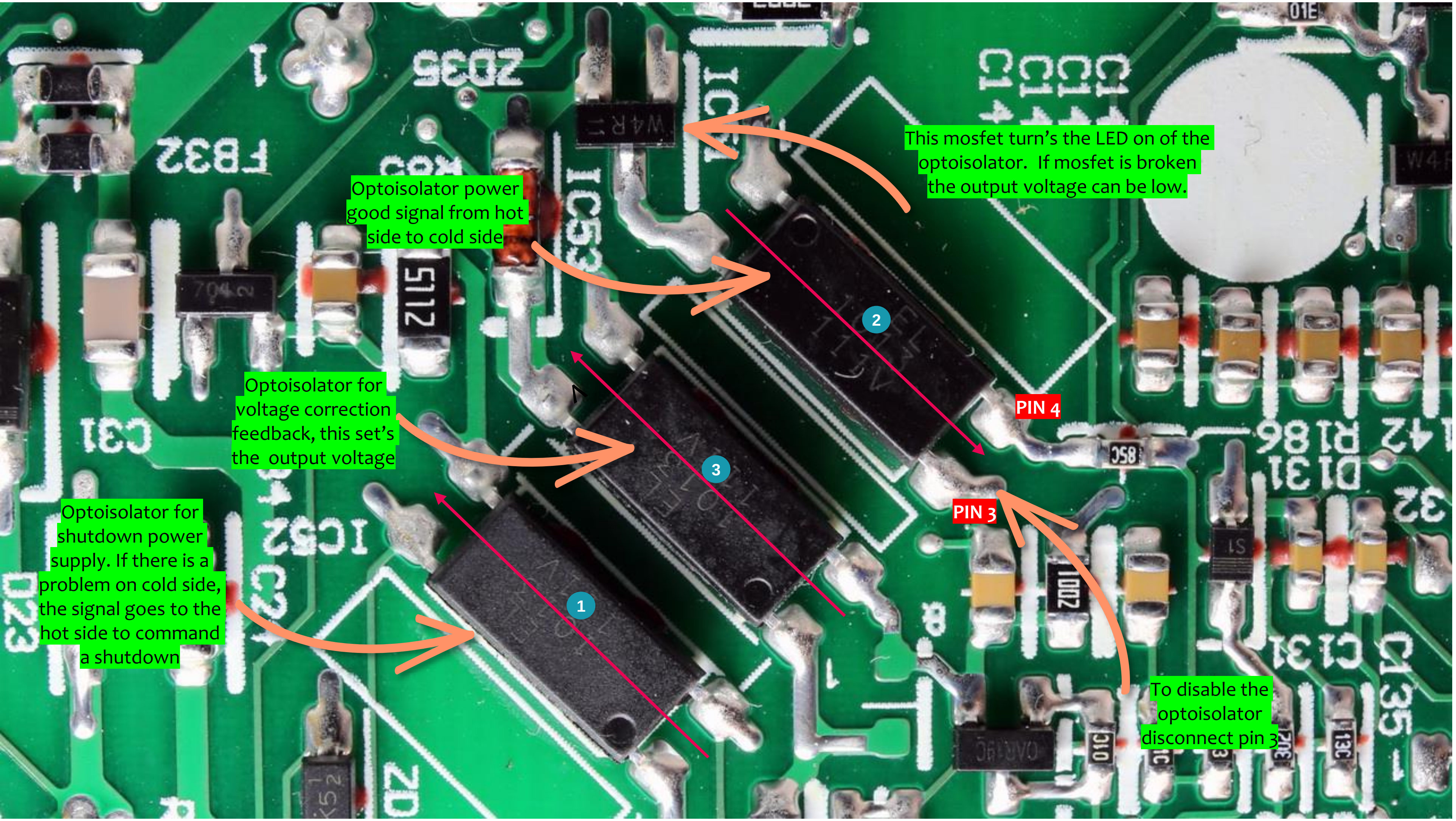
Resistor has continuity then resistor and is low resistance then it is OK

Resistor has no continuity then resistor is blown.

Also check if the Boost Diode and Mosfet are shorted.

Check the APFC / PWM controller IC, because the IC drives these diode and mosfet for the PFC circuit.

Optoisolator for voltage correction feedback



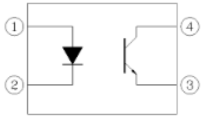
Feedback loop

By using an optoisolator in the feedback loop, the output voltage can be constantly monitored and regulated. Any deviation from the desired voltage will be detected by the optocoupler, allowing the control circuit to make appropriate adjustments.

OUTPUT < +12v check mosfet / optoisolator

If main capacitors voltage is higher then +- 390V check if these optoisolator are ok and not leaking.

Measure the resistance of the collector / emitter junction between pin 3 and 4 of the optoisolator. It should be infinite resistance



Pin Configuration
1. Anode
2. Cathode
3. Emitter
4. Collector